

Nonlinear Solid Mechanics

A COMPILATION OF
NOTES, NOTATIONS, AND
ILLUSTRATIONS

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Abstract

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Acknowledgements

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Chapter 1

Nomenclature

Tensors	
a	zeroth order (scalar)
$\mathbf{A}$	first order (vector) in the reference configuration
A_I	first order (vector) components in the reference configuration
$\mathbf{a}$	first order (vector) in the current configuration
a_i	first order (vector) components in the current configuration
$\mathbf{A}$	second order (matrix) in the reference configuration
$\mathbf{a}$	second order (matrix) in the current configuration
A_{IJ}	second order (matrix) components in the reference configuration
a_{ij}	second order (matrix) components in the current configuration
$\mathbb{A}$	fourth order in the reference configuration
$\mathbb{A}_{IJKL}$	fourth order components in the reference configuration
$\mathbb{a}$	fourth order in the current configuration
$\mathbb{a}_{ijkl}$	fourth order components in the current configuration
$\hat{\mathbf{E}}_I$	unit basis vector in the reference configuration
$\hat{\mathbf{e}}_i$	unit basis vector in the current configuration
O	origin in the reference configuration
o	origin in the current configuration

Motion	
$\mathbf{X}$	position of material particle in reference configuration
$\mathbf{x}$	position of material particle in current configuration
$\mathbf{u}$	displacement vector from $\mathbf{X}$ to $\mathbf{x}$
φ	motion of material particle $\mathbf{X}$
$\varphi_*[\cdot]$	push forward operator
$\varphi_*^{-1}[\cdot]$	pull back operator
$\mathbf{F} = \nabla_0 \varphi$	deformation gradient
$\nabla_0 \mathbf{u}$	displacement gradient in the reference configuration
$\nabla \mathbf{u}$	displacement gradient in the current configuration
$\mathbf{R}$	rotation
$\mathbf{U}$	stretch
Strain	
$\mathbf{C}$	right Cauchy-Green
$\mathbf{b}$	left Cauchy-Green
$\mathbf{E}$	Green-Lagrange
$\mathbf{e}$	Almansi-Euler
$\mathbf{E}_{(m)}$	Seth-Hill
λ_I	principal stretch ratio
ϵ	infinitesimal

Stress	
σ	Cauchy
τ	Kirchhoff
$\mathbf{P}$	Piola-Kirchhoff One
$\mathbf{S}$	Piola-Kirchhoff Two
Energy	
P	pressure
V	volume
$\mathcal{V}$	specific volume
T	temperature
S	entropy
ρ	density
m	mass
E	energy
$\mathcal{E}$	specific energy
F	Helmholtz free energy
G	Gibbs free energy
H	enthalpy

Chapter 2

Introduction

2.1 Purpose and Scope

- The purpose of this manuscript is to memorialize the theories of solid dynamics, thermodynamics, and shock physics.
- The scope is simply to include a listing of variables, equations, and methods used. It is not intended to be a substitute for textbooks on continuum mechanics and the finite element method.

2.2 Mathematical Preliminaries

As defined by Gurtin,¹

- LIN= the set of all tensors
- LIN⁺= the set of all tensors **S** with $\det \mathbf{S} > 0$
- SYM= set of all symmetric tensors
- SKW= set of all skew tensors
- PSYM= set of all positive definite, symmetric tensors
- ORTH= set of all orthogonal tensors
- ORTH⁺= set of all rotations

As defined by Curnier,² for all second-order tensors **S** $\in$ LIN, the **tracor** $\mathbb{T}$ and **deviator** $\mathbb{D}$ are fourth-order tensors are defined as

$$\mathbb{T} \triangleq \frac{1}{3} \delta_{ij} \delta_{kl} \hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j \otimes \hat{\mathbf{e}}_k \otimes \hat{\mathbf{e}}_l \quad (2.1)$$

such that

$$\mathbb{T} : \mathbf{S} = \mathbf{S}_{\text{VOL}} \triangleq \frac{1}{3} \text{tr}(\mathbf{S}) \mathbf{1}, \quad (2.2)$$

and

$$\mathbb{D} \triangleq \left(\frac{1}{2} \delta_{ik} \delta_{jl} + \frac{1}{2} \delta_{il} \delta_{jk} - \frac{1}{3} \delta_{ij} \delta_{kl} \right) \hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j \otimes \hat{\mathbf{e}}_k \otimes \hat{\mathbf{e}}_l \quad (2.3)$$

such that

$$\mathbb{D} : \mathbf{S} = \mathbf{S}_{\text{DEV}} \triangleq \mathbf{S} - \frac{1}{3} \text{tr}(\mathbf{S}) \mathbf{1}. \quad (2.4)$$

Note, there are two other fourth-order tensors often used, the **symmetrizer** $\mathbb{S}$ and the **skewer** $\mathbb{W}$ defined, respectively, as

$$\mathbb{S} \triangleq \frac{1}{2} (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) \hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j \otimes \hat{\mathbf{e}}_k \otimes \hat{\mathbf{e}}_l, \quad (2.5)$$

and

$$\mathbb{W} \triangleq \frac{1}{2} (\delta_{ik} \delta_{jl} - \delta_{il} \delta_{jk}) \hat{\mathbf{e}}_i \otimes \hat{\mathbf{e}}_j \otimes \hat{\mathbf{e}}_k \otimes \hat{\mathbf{e}}_l. \quad (2.6)$$

¹Gurtin ME. An introduction to continuum mechanics. Academic press; 1982, Vol. 158, Jan 12.

²Curnier A. Computational methods in solid mechanics. Springer Science & Business Media; 2012 Dec 6.

Chapter 3

Formulation

3.1 Bodies and Particles

Let body $\mathbf{B}$ be an open, bounded subset of three-dimensional Euclidean space $\mathbb{R}^{n_{\text{sd}}}$ with boundary $\partial\mathbf{B}$. The number of space dimensions, n_{sd} , is assumed equal to three unless stated otherwise. The boundary is decomposed within a particular space dimension into two non-overlapping subdomains, Γ_u and Γ_σ where displacements or tractions are prescribed. Thus $\partial\mathbf{B} = \Gamma_u \cup \Gamma_\sigma$ and $\Gamma_u \cap \Gamma_\sigma = \emptyset$. Set closure is denoted $\bar{\mathbf{B}} = \mathbf{B} \cup \partial\mathbf{B}$. Let the body be composed of a closed set of particles with coordinates $\mathbf{X}$ in the reference configuration. The particle has coordinates $\mathbf{X} = \{X_1, X_2, X_3\}^T$ as measured from some origin O in the dextral, orthonormal inertial reference from $\hat{\mathbf{E}}_I, \hat{\mathbf{E}}_J, \hat{\mathbf{E}}_K$.

3.2 Motion

Let the arbitrary time interval be defined as $\mathbb{T} \triangleq [t_0, T]$.¹ Let the motion, a one parameter family of configurations, $\varphi : \mathcal{B} \times \mathbb{T} \mapsto \mathbb{R}^3$ map the material particle $\mathbf{X}$ into the current configuration $\mathbf{x}$ such that

$$\mathbf{x} = \varphi(\mathbf{X}, t) = \varphi_t(\mathbf{X}). \quad (3.1)$$

A motion evaluated at a particular time $t \in \mathbb{T}$ is referred to as a configuration or placement. To each configuration, we assign a displacement field $\mathbf{u} : \mathcal{B} \mapsto \mathbb{R}^3$ such that

$$\mathbf{u}(\mathbf{X}, t) = \varphi(\mathbf{X}, t) - \mathbf{X} \iff u_i = \varphi_i - \delta_{iJ}X_J. \quad (3.2)$$

Thus, the current configuration φ is simply a function of the original placement $\mathbf{X}$, plus a displacement $\mathbf{u}$, which is **function** of placement $\mathbf{X}$ and time t :

$$\varphi(\mathbf{X}, t) = \mathbf{X} + \mathbf{u}(\mathbf{X}, t) \iff \varphi_i = \delta_{iJ}X_J + u_i. \quad (3.3)$$

It is important to emphasize that the displacement remains a function $\mathbf{u} = \mathbf{u}(\mathbf{X}, t)$ of reference configuration $\mathbf{X}$ and time t . The displacement function $\mathbf{u}$ is not a simple (constant) displacement, which we refer to as “offset” in Section 3.2.2.

¹Note that t_0 , while typically zero, may be any real number.

3.2.1 Linear Motions

In this section, we consider the equation of a line

$$y = f(x) = mx + b, \quad (3.4)$$

as inspiration for writing (3.3) as²

$$\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}) = \underbrace{\mathbf{M}\mathbf{X}}_{\text{linear in } \mathbf{X}} + \underbrace{\mathbf{b}}_{\text{constant}} \stackrel{\text{set}}{=} \mathbf{X} + \mathbf{u}(\mathbf{X}). \quad (3.5)$$

Then the displacement can be written as

$$\mathbf{u}(\mathbf{X}) = \mathbf{M}\mathbf{X} + \mathbf{b} - \mathbf{X}, \quad (3.6)$$

$$= (\mathbf{M} - \mathbf{1})\mathbf{X} + \mathbf{b}. \quad (3.7)$$

In the following sections, we will be exploring the effect of “slope” $\mathbf{M}$ and “intercept” $\mathbf{b}$ on displacements $\mathbf{u}$. To explore some motions, we shall consider the two-dimensional body shown in Fig. 3.1.

The most elementary displacement is homogeneous, when $\mathbf{M} = \mathbf{1}$ and $\mathbf{b} = \mathbf{0}$, then $\mathbf{u} = \mathbf{0}$. In this case $\boldsymbol{\varphi} = \mathbf{X}$ and the motion is trivial (and uninteresting). Nonetheless, this is a simple first case to consider.

3.2.2 Offset

Next, consider $\mathbf{b} \neq \mathbf{0}$, while retaining $\mathbf{M} = \mathbf{1}$. For the concrete example in Fig. (3.2), let $\mathbf{b} = \langle b_1, b_2 \rangle^\top = \langle 3, 2 \rangle^\top$. In this case, we see the placement $\boldsymbol{\varphi}$ moves right and up on the page, relative to the reference configuration $\mathbf{X}$, by an amount of 3 and 2, respectively.

²For the time being, we will not be concerned with time t , so we eliminate it from our notation for the sake of brevity.

Figure 3.1: Reference configuration of a body defined by particles with $X_1 \in [-1, 1]$ and $X_2 \in [-1, 1]$. With the continuum body is composed of an infinite number of particles, only nine particles are shown for the sake of simplicity. Eight of the nine particles shown are on the boundary of the body, and a single particle at the geometric center is the ninth particle.

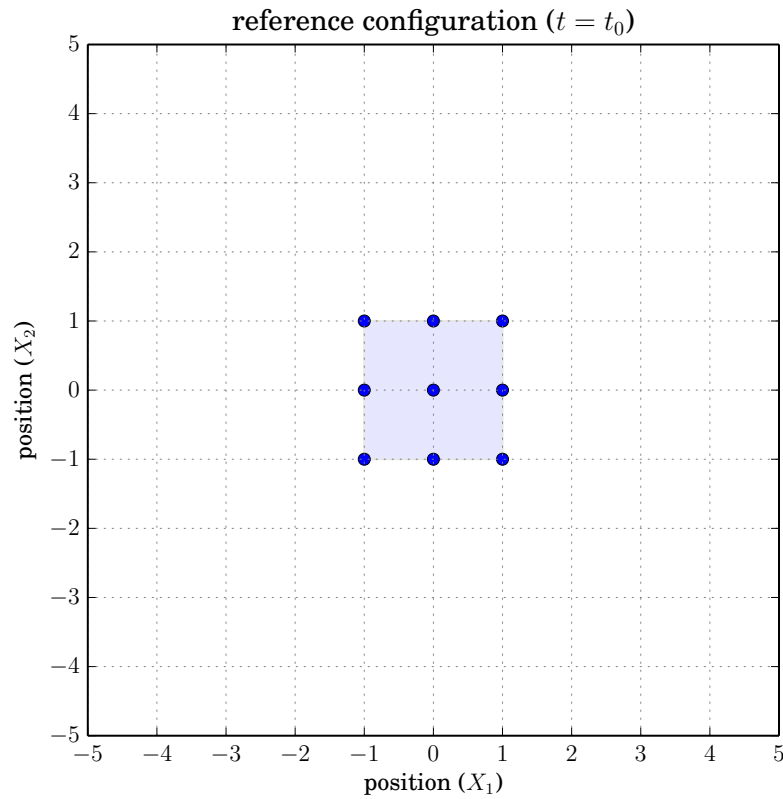
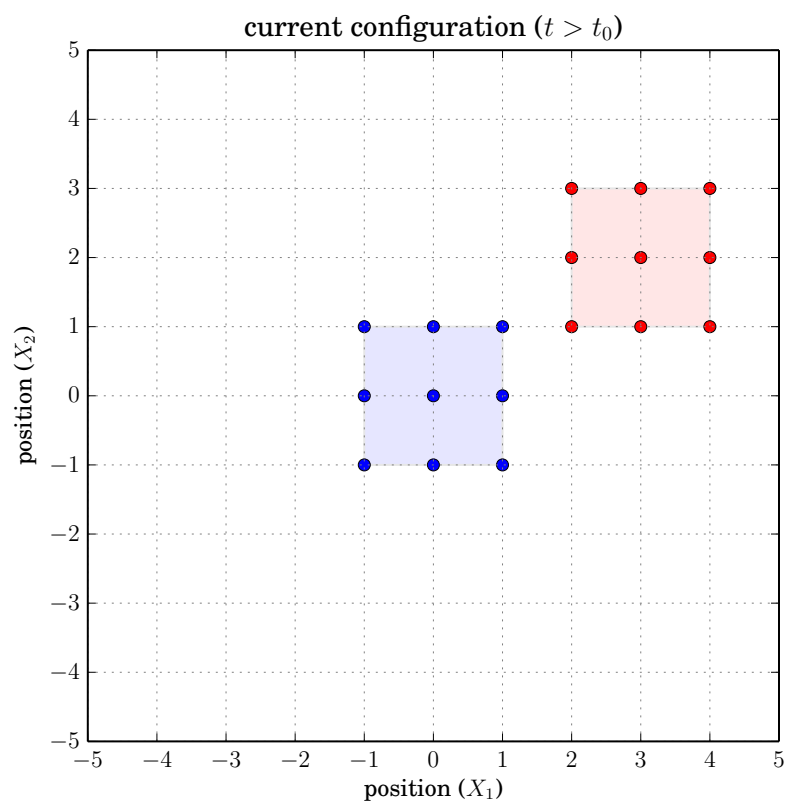


Figure 3.2: Current configuration shown in red, with reference configuration shown in blue and retained for context.



3.2.3 Stretch

Consider now $\mathbf{M} = \lambda \mathbf{1}$ and $\mathbf{b} = \mathbf{0}$. This describes **volumetric** response, also known as a **dilatational** response. The parameter $\lambda \in \mathbb{R}$ causes the current configuration to change volume as

- $\lambda > 0$ expansion,
- $\lambda = 0$ identity,
- $\lambda < 0$ contraction.

Notice that λ describes the “slope” of the mapping from reference to current configurations. In the case that $\lambda \rightarrow +\infty$, the volume grows infinite. In the case that $\lambda \rightarrow -\infty$, the volume grows to zero. For concreteness, consider $\lambda = 2$. The volumetric expansion is shown in Fig. 3.3.

The area in the reference configuration was $2 \times 2 = 4$, whereas the area in the current configuration was $4 \times 4 = 16$.

Next consider a slightly different form of $\mathbf{M}$, such that

$$\mathbf{M} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}. \quad (3.8)$$

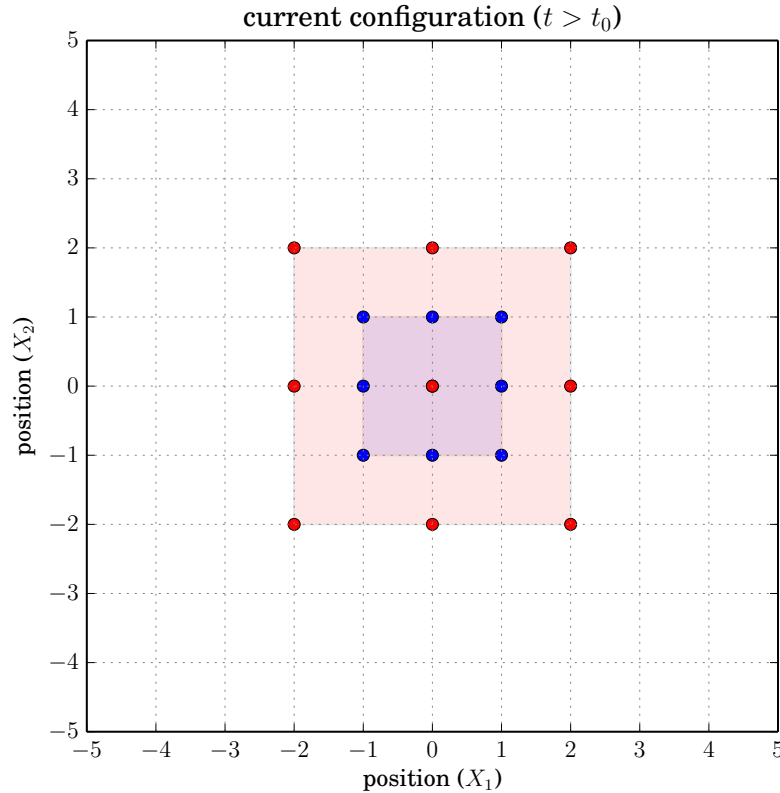
We keep the offset as $\mathbf{b} = \mathbf{0}$.

For concreteness, consider $\lambda_1 = 2$ and $\lambda_2 = \frac{1}{2}$. The combined expansion and compression is shown in Fig. 3.4.

The area in the reference configuration was $2 \times 2 = 4$, whereas the area in the current configuration was $4 \times 1 = 4$. Out of coincidence, we have chosen stretches along the two axes such that the area does not change although the dimensions and thus shape have changed. This is an example of an **isochoric** motion. The roots *iso-* (*equal*) and *choro-* (*space*) describe a process from initial state 0 to final state 1 where the volume remains constant.

Now, the motion as we have described it, doesn’t yet show the *path* taken between the reference and current configuration. To get that detail, we would need to introduce the time parameter t , which we have suppressed for now. However, if we were to say the time interval is simply two points t_0 and t_1 , we can create (the rather course) path between these two states 0 and 1, and connect with the isochoric process *PV* diagram in thermodynamics, shown in Fig. 3.5.

Figure 3.3: Current configuration shown in red, with reference configuration shown in blue and retained for context.



We have yet to introduce and define pressure P , so this connection may be tenuous at the moment. Nonetheless, we introduce this connection now to motivate the single idea that isochoric motions will play an important role in our later work.

3.2.4 Rotation

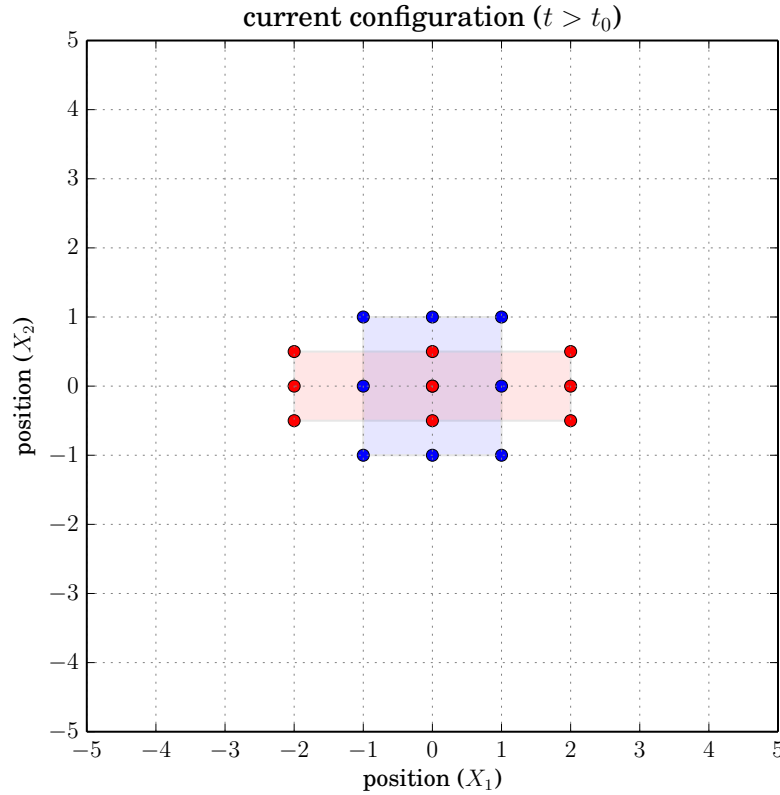
3.3 Deformation Gradient

To each configuration, we define a deformation gradient $\mathbf{F} : \mathcal{B} \times \mathbb{T} \mapsto \mathbb{M}_+^3$ such that

$$\mathbf{F} \triangleq \nabla_0 \varphi \iff F_{iJ} \triangleq \frac{\partial \varphi_i}{\partial X_J}. \quad (3.9)$$

Remarks:

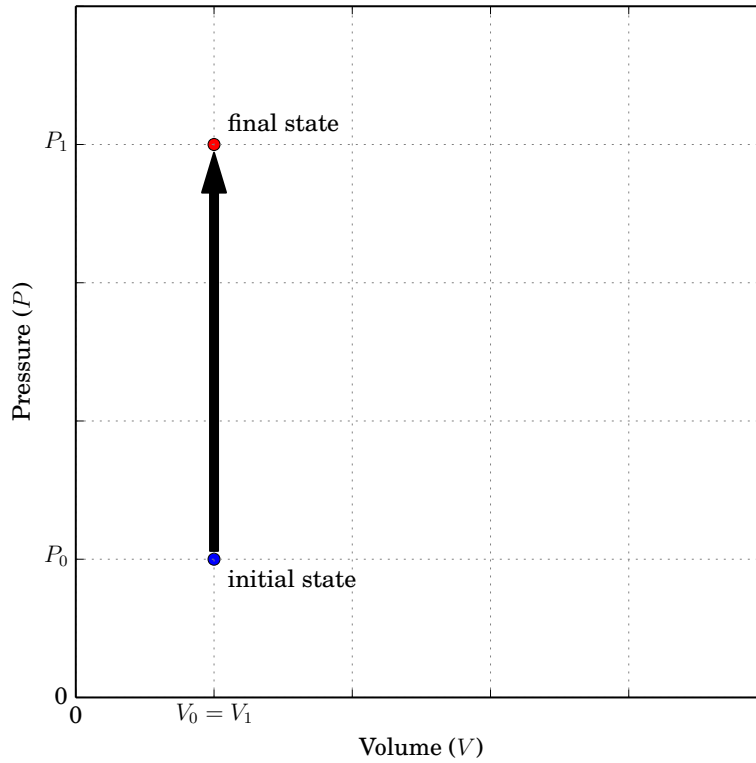
Figure 3.4: Current configuration shown in red, with reference configuration shown in blue and retained for context.



- Lower case indices represent components of tensors in the current configuration whereas upper case indices represent components of tensors in the reference configuration. Real, square matrices of dimension three and positive determinants are denoted $\mathbb{M}_+^3$.
- We define gradients taken in the reference configuration as $\nabla_0(\cdot) = \partial(\cdot)/\partial X_J$ to distinguish from gradients taken in the current configuration, denoted simply as $\nabla(\cdot) = \partial(\cdot)/\partial x_j$.
- All configurations must be admissible in the sense that the Jacobian of the deformation J must be positive

$$J \triangleq \det \mathbf{F} > 0. \quad (3.10)$$

Figure 3.5: Isochoric process from initial state (reference configuration) 0 to final state (current configuration) 1.



This requirement keeps the deformations from mapping the body to a single, infinitesimally small point ($J = 0$) or turning the body inside-out ($J < 0$).

From the displacement field in (3.2), we see that the relationship between the displacement gradient and the deformation gradient is given by

$$\nabla_0 \mathbf{u} = \nabla_0 \boldsymbol{\varphi} - \mathbf{1} = \mathbf{F} - \mathbf{1} \iff \frac{\partial u_i}{\partial X_J} = \frac{\partial \varphi_i}{\partial X_J} - \delta_{iJ}. \quad (3.11)$$

3.4 Strain

3.4.1 Right Cauchy-Green

A standard strain measure is the right Cauchy-Green strain tensor

$$\mathbf{C} \triangleq \mathbf{F}^\top \mathbf{F} \iff C_{IJ} \triangleq F_{Ii} F_{iJ}. \quad (3.12)$$

The right Cauchy-Green strain tensor $\mathbf{C}$ gets its name from the fact that the deformation gradient $\mathbf{F}$ sits to the right. Note that this strain tensor is defined in the reference configuration. The right Cauchy-Green strain tensor can be written explicitly as reference gradients of the motion $\boldsymbol{\varphi}$ or displacement $\mathbf{u}$:

$$\mathbf{C} \stackrel{(3.9)}{=} (\nabla_0 \boldsymbol{\varphi})^\top \nabla_0 \boldsymbol{\varphi} \stackrel{(3.11)}{=} (\mathbf{1} + \nabla_0 \mathbf{u})^\top (\mathbf{1} + \nabla_0 \mathbf{u}) \quad (3.13)$$

$$= \mathbf{1} + \nabla_0 \mathbf{u} + (\nabla_0 \mathbf{u})^\top + (\nabla_0 \mathbf{u})^\top \nabla_0 \mathbf{u} \quad (3.14)$$

3.4.2 Left Cauchy-Green

In contrast, a left Cauchy-Green strain tensor $\mathbf{b}$ can be defined with the deformation gradient $\mathbf{F}$ sitting on the left.

$$\mathbf{b} \triangleq \mathbf{F} \mathbf{F}^\top \iff b_{ij} \triangleq F_{iJ} F_{jJ}. \quad (3.15)$$

Note that this strain measure is defined in the current configuration.

3.4.3 Green-Lagrange

The Green-Lagrange strain tensor,

$$\mathbf{E} \triangleq \frac{1}{2}(\mathbf{C} - \mathbf{1}) \iff E_{IJ} \triangleq \frac{1}{2}(C_{IJ} - \delta_{IJ}), \quad (3.16)$$

is closely related to the right Cauchy-Green strain tensor and is often used in defining constitutive law relationships because the measure, when linearized about the reference configuration, coincides with the small strain tensor of linear deformation elasticity, denoted $\boldsymbol{\epsilon}$ and defined in (to come). This relationship can be seen as follows:

$$2\mathbf{E} \stackrel{(3.14)}{=} \mathbf{1} + \nabla_0 \mathbf{u} + (\nabla_0 \mathbf{u})^\top + (\nabla_0 \mathbf{u})^\top \nabla_0 \mathbf{u} - \mathbf{1}, \quad (3.17)$$

$$2\mathbf{E}_{\text{LIN}} = \nabla_0 \mathbf{u} + (\nabla_0 \mathbf{u})^\top, \quad (3.18)$$

where the higher-order (quadratic) term in (3.17) is set to zero to achieve (3.18).

3.4.4 Almansi-Euler

The Almansi-Euler strain tensor,

$$\mathbf{e} \triangleq \frac{1}{2}(\mathbf{1} - \mathbf{b}^{-1}) \iff \mathbf{e}_{ij} \triangleq \frac{1}{2}(\delta_{ij} - (\mathbf{b}_{ij})^{-1}), \quad (3.19)$$

can likewise be used to approximate the small strain tensor $\boldsymbol{\epsilon}$ by combining (3.15) and (3.9) as follows:

$$\begin{aligned} 2\mathbf{e} &= \mathbf{1} - [\nabla_0 \boldsymbol{\varphi} (\nabla_0 \boldsymbol{\varphi})^\top]^{-1} \\ &= \delta_{ij} - \left[\frac{\partial \varphi_i}{\partial X_J} \frac{\partial \varphi_j}{\partial X_J} \right]^{-1} = \delta_{ij} - \left[\frac{\partial X_J}{\partial \varphi_i} \frac{\partial X_J}{\partial \varphi_j} \right] \\ &\stackrel{(3.2)}{=} \mathbf{1} - [(\mathbf{1} - \nabla \mathbf{u})(\mathbf{1} - \nabla \mathbf{u})^\top] \\ &= \nabla \mathbf{u} + \nabla \mathbf{u}^\top - \nabla \mathbf{u} \nabla \mathbf{u}^\top \end{aligned} \quad (3.20)$$

$$2\mathbf{e}_{\text{LIN}} = \nabla \mathbf{u} + (\nabla \mathbf{u})^\top, \quad (3.21)$$

where higher-order (quadratic) term in (3.20) is set to zero to achieve (3.21).

3.4.5 Small Strain Theory

When displacement gradients are small in the reference configuration

$$\nabla_0 \mathbf{u} \ll 1, \quad (3.22)$$

or in the current configuration,

$$\nabla \mathbf{u} \ll 1, \quad (3.23)$$

respectively, the nonlinear gradient terms are negligible and the finite strain theory simplifies to small strain theory, which occurs when finite strain measures are linearized to obtain $\mathbf{E}_{\text{LIN}}$ and $\mathbf{e}_{\text{LIN}}$ in the previous sections.

Note that we have restricted the gradients of displacement, and not the displacement $\mathbf{u}$ itself. Thus, displacements between the reference and current configurations can be large (finite), but the gradients of the displacement either in the reference or current configuration are small.

Section 3.4.8 will demonstrate that the small strain tensors are not suitable to describe motion that contains finite rotation. This makes sense because, in finite rotation, gradients of displacement are large, not small. To adequately describe motion that includes finite rotation, the fully nonlinear strain measured must be used.

3.4.6 Infinitesimal Strain Theory

If we further restrict the small strain theory such that the displacement is small compared to unity,

$$\mathbf{u} \ll 1, \quad (3.24)$$

the infinitesimal strain theory is obtained, which has no distinction between Lagrangian and Eulerian strain tensors.

In this case, the two small strain tensors, $\mathbf{E}_{\text{LIN}}$ and $\mathbf{e}_{\text{LIN}}$ converge to a single definition of strain, called the infinitesimal strain tensor $\boldsymbol{\epsilon}$, defined as

$$\boldsymbol{\epsilon} \triangleq \frac{1}{2}(\nabla \mathbf{u} + (\nabla \mathbf{u})^\top). \quad (3.25)$$

Note that the $\nabla_0(\cdot)$ notation has been dropped since the distinction between the reference and current configurations is nonexistent. Also, note that the factor of $\frac{1}{2}$ appears because it then follows that the infinitesimal strain is simply the symmetric part of the displacement gradient,

$$\boldsymbol{\epsilon} = \text{SYM}(\nabla \mathbf{u}). \quad (3.26)$$

Finally, note that the finite Lagrangian and Eulerian strain tensors were defined with the factor of $\frac{1}{2}$ so that their expressions, once linearized and subject to a small displacement assumption, simplify to exactly the infinitesimal strain tensor $\boldsymbol{\epsilon}$.

3.4.7 Seth-Hill Strain Tensor Family

We return to finite strain definitions, and show that the Lagrangian and Eulerian strain tensors described above are two special cases of a family of finite strain tensors.

Seth and Hill showed that the Green-Lagrange and Almansi-Euler strain tensors are special cases of the so-called Seth-Hill family of strain measures, defined as

$$\mathbf{E}_{(m)} \triangleq \begin{cases} \frac{1}{m} (\mathbf{U}^m - \mathbf{1}) & \text{for } m \neq 0, \\ \ln \mathbf{U} & \text{for } m = 0. \end{cases} \quad (3.27)$$

The principal stretches $\lambda_I = \ell_k/L_0$, $0 < \lambda_I < \infty$, allow the strain measure $\mathbf{E}_{(m)}$ to be

written as principal strains, as a function of principal stretch, $f(\lambda_I)$:

$$\begin{aligned} \mathbf{E}_{(m)} &= \sum_{I=1}^3 f(\lambda_I) \hat{\mathbf{E}}_I \otimes \hat{\mathbf{E}}_I \iff \\ \mathbf{E}_{(m)IJ} &= f(\lambda_1) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + f(\lambda_2) \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + f(\lambda_3) \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \end{aligned} \quad (3.28)$$

where

$$f(\lambda_I) \triangleq \begin{cases} \frac{1}{m} (\lambda_I^m - 1) & \text{for } m \neq 0, \\ \ln(\lambda_I) & \text{for } m = 0. \end{cases} \quad (3.29)$$

For integer values of $m \in [-2, 2]$, five common strain measures result, listed in Table 3.1, in their three-dimensional and one-dimensional forms.

As will be seen in section (to come),

- The Green-Lagrange strain tensor will be the conjugate of the second Piola-Kirchhoff stress tensor,
- The deformation gradient will be the conjugate of the first Piola-Kirchhoff stress tensor,
- The engineering (Biot) strain tensor will be the conjugate of the Jaumann (Biot) stress tensor,
- The Almansi-Euler strain tensor will be the conjugate of the weighted convected stress tensor.

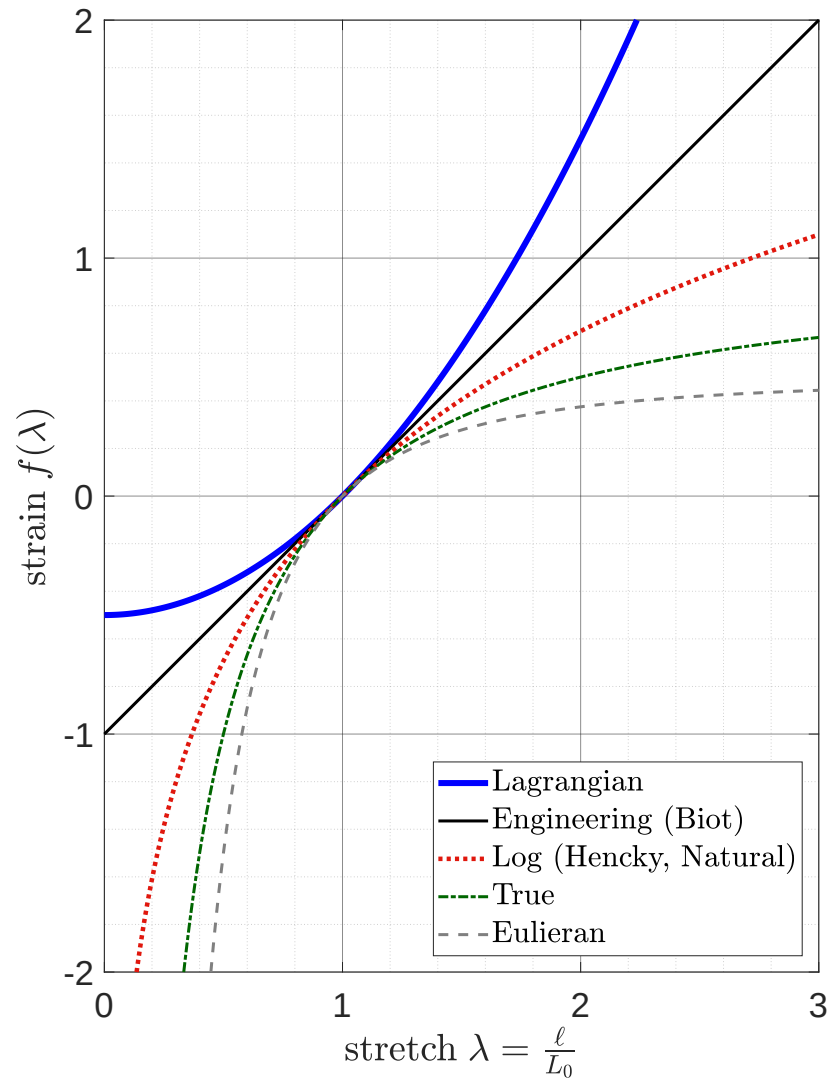
m	Name	3D	1D
2	Green-Lagrange	$\mathbf{E}_{(2)} = \frac{1}{2} (\mathbf{U}^2 - \mathbf{1})$	$E_{\text{LAG}} = \frac{1}{2} \left[\left(\frac{\ell}{L_0} \right)^2 - 1 \right]$
1	Engineering (Biot)	$\mathbf{E}_{(1)} = \mathbf{U} - \mathbf{1}$	$E_{\text{ENG}} = \frac{\ell - L_0}{L_0}$
0	Log (Hencky, Natural)	$\mathbf{E}_{(0)} = \ln \mathbf{U}$	$E_{\text{LOG}} = \ln \left(\frac{\ell}{L_0} \right)$
-1	True	$\mathbf{E}_{(-1)} = \mathbf{1} - \mathbf{U}^{-1}$	$E_{\text{TRUE}} = \frac{\ell - L_0}{\ell}$
-2	Almansi-Euler	$\mathbf{E}_{(-2)} = \frac{1}{2} (\mathbf{1} - \mathbf{U}^{-2})$	$E_{\text{EUL}} = \frac{1}{2} \left[1 - \left(\frac{L_0}{\ell} \right)^2 \right]$

Table 3.1: Table of strains obtained from the Seth-Hill family.

Figure 3.6 illustrates the one-dimensional strains as a function of stretch ratio $\lambda = \ell/L_0$. The figure illustrates several results:

- For small stretches, $\ell \approx L_0$,
 - The stretch ratio is near unity, $\lambda \approx 1$,
 - The strain values are small, $f(\lambda) \approx 0$,
 - The tangent of the strains with respect to the stretch ratio is near unity, $df/d\lambda \approx 1$.
- For elongations, $\lambda > 1$,
 - The strain monotonically increases since $df/d\lambda > 0$ when $\lambda > 0$.
- For extreme compressions, $\lambda \approx 0$,
 - The Green-Lagrange strain goes to a value of $-\frac{1}{2}$,
 - The engineering (Biot) strain tensor goes to a value of -1 ,
 - The Log, true, and Eulerian strains tend to $-\infty$.
- The engineering (Biot) strain is a linear function of stretch λ ; all other measures are nonlinear functions of stretch λ .

Figure 3.6: One dimensional strain as a function of stretch ratio.



3.4.8 The Linearized Green-Lagrange Strain Tensor Creates Nonzero Strains under Rigid Body Rotations

Because it takes on nonzero values under finite rotation, the linearized strain tensor should not be used for geometrically nonlinear analysis. These nonzero values are completely artificial and strictly an artifact of using a linear strain definition with geometrically nonlinear motions. This result is shown as follows:

Let $\mathbf{R} = \mathbf{R}(t)$ be a two-dimensional, rigid body rotation parameterized by time t and scaled by constant $\omega = 2\pi$ radians per second. Then, the motion $\boldsymbol{\varphi}$ of a body can be written as

$$\boldsymbol{\varphi}(\mathbf{X}, t) = \mathbf{R}(t)\mathbf{X} \iff \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = \begin{bmatrix} \cos \omega t & \sin \omega t \\ -\sin \omega t & \cos \omega t \end{bmatrix} \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix}. \quad (3.30)$$

Then the deformation gradient $\mathbf{F} = \mathbf{F}(t)$ is a function of time alone,

$$\mathbf{F}(t) \iff F_{iJ} = \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix} = \begin{bmatrix} \cos \omega t & \sin \omega t \\ -\sin \omega t & \cos \omega t \end{bmatrix}. \quad (3.31)$$

The linearized strain tensor is found to be

$$2\mathbf{E}_{\text{LIN}} = \boldsymbol{\nabla}_0 \mathbf{u} + (\boldsymbol{\nabla}_0 \mathbf{u})^T, \quad (3.32)$$

$$= \mathbf{F} - \mathbf{1} + (\mathbf{F} - \mathbf{1})^T, \quad (3.33)$$

$$= F_{iJ} - \delta_{iJ} + F_{Ji} - \delta_{Ji}, \quad (3.34)$$

$$= \begin{bmatrix} \cos \omega t & \sin \omega t \\ -\sin \omega t & \cos \omega t \end{bmatrix} + \begin{bmatrix} \cos \omega t & -\sin \omega t \\ \sin \omega t & \cos \omega t \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad (3.35)$$

$$= 2(\cos \omega t - 1)\mathbf{1}. \quad (3.36)$$

Now, for small angles, $\omega t \approx 0$, which is for small deviations $x_1 \approx X_1$, $x_2 \approx X_2$, then $\mathbf{E}_{\text{LIN}} = \mathbf{0}$ for rigid body rotations. However, for arbitrary finite angles, $\omega t \neq 0$, and the linearized strain tensor $\mathbf{E}_{\text{LIN}}$ reports nonzero strain for rigid body rotations, which is nonsensical.

A correct strain tensor will report zero strain for rigid body rotations. One such strain tensor is the fully nonlinear Green-Lagrange strain tensor. This result is shown as follows:

$$2\mathbf{E} = \mathbf{F}^T \mathbf{F} - \mathbf{1} \quad (3.37)$$

$$= \begin{bmatrix} \cos \omega t & -\sin \omega t \\ \sin \omega t & \cos \omega t \end{bmatrix} \begin{bmatrix} \cos \omega t & \sin \omega t \\ -\sin \omega t & \cos \omega t \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{0}. \quad (3.38)$$

3.5 Stress

3.6 Hyperelasticity

3.7 Balance Laws

In this section, we posit the **Master Balance Law** in general form. Also called “conservation of anything,” this description assumes an arbitrary control volume $V \in \mathbb{R}^3$ with boundary ∂V . What follows is the so-called Eulerian formulation, illustrated in Fig. 3.7, which assumes the control volume, while arbitrary in its placement in space, is **fixed** in space.³

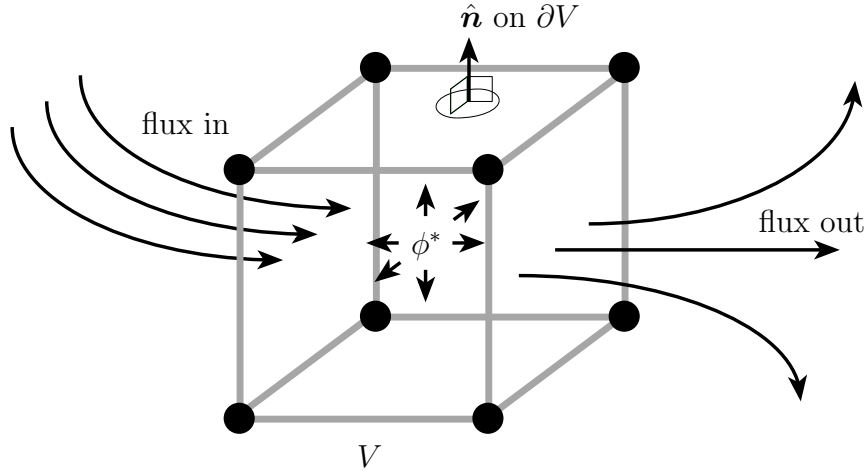


Figure 3.7: Concept of a fixed control volume of the Eulerian formulation.

Now, let

- $\phi(\mathbf{x}, t)$ be the description of some physical quantity per unit volume, which can in general be a scalar, vector, or tensor quantity,
- $\phi^*(\mathbf{x}, t)$ be the generation (depletion) *rate* of the physical quantity per unit volume,
- $\hat{\phi}(\mathbf{x}, t)$ be the physical quantity per unit volume that flows across the boundary with rate (velocity) $\mathbf{v}(\mathbf{x}, t)$, and
- ∂V be the boundary of the control volume, with outward pointing surface normal unit vector $\hat{\mathbf{n}}(\mathbf{x}, t)$.

³The connection to the Lagrangian formulation, in which the control is not fixed in space, but instead, moves through space as the body to which it is attached moves through space, is examined in the sequel.

The time rate of change of the quantity $\phi(\mathbf{x}, t)$ within the differential control volume dV is then equal to

- **generation**—the rate at which the quantity is generated (or depleted) within the control volume by the point source (point sink), plus
- **flux**—the rate at which the quantity is either carried into (or out of) the volume across the volume's differential surface area dA .

These words, stated mathematically, are

$$\underbrace{\frac{d}{dt} \int_V \phi(\mathbf{x}, t) dV}_{\text{rate}} = \underbrace{\int_V \phi^*(\mathbf{x}, t) dV}_{\text{source}} - \underbrace{\int_{\partial V} [\hat{\phi}(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)] \cdot \hat{\mathbf{n}} dA}_{\text{flux}}, \quad (3.39)$$

which is the general form of the **Master Balance Law**. We adopt the following sign convention to impart consistency:

- For positive time rate of change of the quantity $\phi(\mathbf{x}, t)$, the generation $\phi^*(\mathbf{x}, t)$ is trivially positive. For quantity $\hat{\phi}(\mathbf{x}, t)$ carried *into* the control volume, the inner product of $\mathbf{v}$ with $\hat{\mathbf{n}}$ is negative, thus a negative outside of the flux integral is used.
- Likewise, for a negative time rate of change of quantity $\phi(\mathbf{x}, t)$, the generation $\phi^*(\mathbf{x}, t)$ is negative. And, for the quantity $\hat{\phi}(\mathbf{x}, t)$ carried *out of* the control volume, the inner product of $\mathbf{v}$ with $\hat{\mathbf{n}}$ is positive, thus the negative outside of the flux integral provides sign consistency.

3.7.1 Balance Law Metaphor

Here we present the Master Balance Law by way of metaphor. Let the volume V be the geographic region of Oregon, with boundary ∂V that touches the state borders of Washington, Idaho, Nevada, and California in the United States.⁴ Let ϕ be the number of people living in the state of Oregon. The change in population $\Delta\phi$ over some time interval Δt is equal to the net generation rate ϕ^* (number of births less the number of deaths in the time interval), plus the net population flux rate $\hat{\phi}\mathbf{v}$ (the number of people moving to Oregon less the number of people moving from Oregon within the time interval).

⁴In this oversimplified metaphor, we assume flux does not occur via the western seaboard nor via the air.

3.7.2 Conservation of Mass

Let us now apply the Master Balance Law to the physical quantity of mass, and develop a relationship for conservation of mass, a concept where mass is neither created nor destroyed. When we speak of the physical quantity of mass per unit volume, we speak of density $\rho(\mathbf{x}, t)$. Thus, we substitute $\rho(\mathbf{x}, t)$ for $\phi(\mathbf{x}, t)$ in the foregoing Master Balance Law.

For mass to be conserved, mass within a control volume is neither created nor destroyed, thus the source term is identically zero. Thus, increases (decreases) in the mass rate term are produced only by flux of mass in through (out from) the boundary. The idea, stated mathematically, is

$$\underbrace{\frac{d}{dt} \int_V \rho(\mathbf{x}, t) dV}_{\text{rate}} = \underbrace{\int_V \rho^*(\mathbf{x}, t) dV}_{\text{source} = 0} - \underbrace{\int_{\partial V} [\rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)] \cdot \hat{\mathbf{n}}(\mathbf{x}, t) dA}_{\text{flux}}. \quad (3.40)$$

The Divergence Theorem is used to write the flux term surface integral as a volume integral, thus

$$\underbrace{\frac{d}{dt} \int_V \rho(\mathbf{x}, t) dV}_{\text{rate}} = 0 - \underbrace{\int_V \nabla \cdot [\rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)] dV}_{\text{flux}}. \quad (3.41)$$

Because V is fixed in an inertial frame, the foregoing equation can be rewritten with the rate time derivative inside the integral, thus

$$\int_V \left(\frac{\partial \rho(\mathbf{x}, t)}{\partial t} + \nabla \cdot [\rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)] \right) dV = 0. \quad (3.42)$$

Finally, arbitrariness of B_t requires that the integrand is identically zero, thus

$$\frac{\partial \rho(\mathbf{x}, t)}{\partial t} + \nabla \cdot [\rho(\mathbf{x}, t) \mathbf{v}(\mathbf{x}, t)] = 0 \iff \rho_{,t} + (\rho v_i)_{,i} = 0, \quad (3.43)$$

which is the **continuity equation**. Some texts will recast the continuity equation by defining a **material derivative** of density as follows: Since $\rho = \rho(\mathbf{x}, t)$, then

$$\dot{\rho} = \frac{\partial \rho}{\partial \mathbf{x}} \frac{\partial \mathbf{x}}{\partial t} + \frac{\partial \rho}{\partial t} \triangleq \frac{D\rho}{Dt}. \quad (3.44)$$

Expansion of (3.43) then gives

$$\frac{\partial \rho}{\partial t} + \nabla \rho \cdot \mathbf{v} + \rho \nabla \cdot \mathbf{v} = \underbrace{\frac{\partial \rho}{\partial t} + \frac{\partial \rho}{\partial \mathbf{x}} \frac{\partial \mathbf{x}}{\partial t}}_{\triangleq \dot{\rho}} + \rho \nabla \cdot \mathbf{v} = 0, \quad (3.45)$$

or simply

$$\dot{\rho} + \rho \nabla \cdot \mathbf{v} = 0. \quad (3.46)$$

3.7.3 Conservation of Linear Momentum

3.7.4 Conservation of Angular Momentum

3.7.5 Conservation of Energy

3.8 Elastodynamics

3.8.1 Nonlinear Elastodynamics

The bulk modulus B , a measure of local compressibility of a substance, is defined as the ratio of infinitesimal change in pressure dP caused by an infinitesimal change in volume dV , at a given volume V

$$B \triangleq -V \frac{dP}{dV}. \quad (3.47)$$

We require $B > 0$, and since $V > 0$, then $dP/dV < 0$. This results in behaviors observed physically, namely that, all else being equal,

- increase in pressure causes decrease in volume, and vice versa, $\uparrow P \iff \downarrow V$, and
- increase in volume causes decrease in pressure, and vice versa, $\uparrow V \iff \downarrow P$.

Using the definition of density $\rho \triangleq m/V$,

$$\frac{d\rho}{dV} = \frac{-m}{V^2}, \quad (3.48)$$

and

$$\frac{dP}{dV} = \frac{dP}{d\rho} \frac{d\rho}{dV}, \quad (3.49)$$

the bulk modulus B may also be written as

$$B = \rho \frac{dP}{d\rho}. \quad (3.50)$$

Since $B > 0$ and $\rho > 0$, then $dP/d\rho > 0$. This result physically means that, all else being equal,

- increases in pressure cause increases in density (and vice versa), and
- increases in density cause increases in pressure (and vice versa).

3.8.2 Linear Elastodynamics

Linear elastodynamics begins with nonlinear elastodynamics, and adds progressively simplifying (and thus, restrictive) assumptions. The first assumption will be regarding linearity of displacement gradients.

• **Assumption 1:** Second-order and higher displacement gradients are negligible, which restricts the stress-response function to be a special case of the fully general stress response function

$$\mathbf{P} = \mathbf{P}(\mathbf{F}). \quad (3.51)$$

Rewriting the foregoing stress-response function in terms of the displacement gradient,

$$\mathbf{P}(\mathbf{F}) = \mathbf{P}(\mathbf{1} + \nabla_0 \mathbf{u}), \quad (3.52)$$

gives an expression that can be cast in terms of a Taylor series expansion,

$$f(x_0 + \Delta x) = f(x_0) + f'(x_0)\Delta x + \frac{f''(x_0)}{2!}(\Delta x)^2 + \text{H.O.T.}, \quad (3.53)$$

$$= \sum_{j=0}^{\infty} \frac{f^{(j)}(x_0)}{j!} (\Delta x)^j. \quad (3.54)$$

Then

$$\mathbf{P}(\mathbf{1} + \nabla_0 \mathbf{u}) = \mathbf{P}(\mathbf{1}) + \frac{\partial \mathbf{P}(\mathbf{1})}{\partial \mathbf{F}} \nabla_0 \mathbf{u} + \mathcal{O}[(\nabla_0 \mathbf{u})^2]. \quad (3.55)$$

Next, we make another assumption,

• **Assumption 2:** The initial state is stress free, thus $\mathbf{P}(\mathbf{1}) = \mathbf{0}$.

The fourth-order elasticity tensor $\mathbb{C}$ is defined as,

$$\mathbb{C} \triangleq \frac{\partial \mathbf{P}(\mathbf{1})}{\partial \mathbf{F}}. \quad (3.56)$$

Symmetry of the elasticity tensor provides

$$\mathbb{C} : \nabla_0 \mathbf{u} = \mathbb{C} : \text{SYM}(\nabla_0 \mathbf{u}). \quad (3.57)$$

Finally,

$$\text{SYM}(\nabla_0 \mathbf{u}) = \frac{1}{2}(\nabla_0 \mathbf{u} + \nabla_0 \mathbf{u}^\top) = \boldsymbol{\epsilon}. \quad (3.58)$$

These results give

$$\mathbf{P}(\mathbf{1} + \nabla_0 \mathbf{u}) = \mathbf{0} + \mathbb{C} : \boldsymbol{\epsilon} + \text{H.O.T.}, \quad (3.59)$$

or simply

$$\mathbf{P}(\mathbf{1} + \nabla_0 \mathbf{u}) \stackrel{(\text{LIN})}{=} \mathbf{P}_{\text{LIN}}(\boldsymbol{\epsilon}) = \mathbb{C} : \boldsymbol{\epsilon}. \quad (3.60)$$

Next, we can make another limiting assumption,

- **Assumption 3:** The material is **isotropic**,

which allows the stress-reponse function to be written in terms of two Lamé parameters λ and μ as

$$\mathbf{P}_{\text{LIN}}(\boldsymbol{\epsilon}) = \lambda \operatorname{tr}(\boldsymbol{\epsilon}) \mathbf{1} + 2 \mu \boldsymbol{\epsilon}. \quad (3.61)$$

Next, we make another assumption,

- **Assumption 4:** The material is **homogeneous**,

thus ρ_0 , λ , and μ are constants with respect to the position $\mathbf{X}$ in the body.

In the conservation of momentum equation, the divergence of the stress tensor appears. Thus, we find

$$\nabla_0 \cdot \mathbf{P}_{\text{LIN}}(\boldsymbol{\epsilon}) = \nabla_0 \cdot (\lambda \operatorname{tr}(\boldsymbol{\epsilon}) \mathbf{1} + 2 \mu \boldsymbol{\epsilon}) \quad (3.62)$$

$$= \lambda \nabla_0 (\nabla_0 \cdot \mathbf{u}) + \mu [\nabla_0 \cdot (\nabla_0 \mathbf{u}) + \nabla_0 (\nabla_0 \cdot \mathbf{u})], \quad (3.63)$$

$$= (\lambda + \mu) \nabla_0 (\nabla_0 \cdot \mathbf{u}) + \mu \nabla_0 \cdot (\nabla_0 \mathbf{u}). \quad (3.64)$$

Now, since

$$\nabla_0 \cdot (\nabla_0 \mathbf{u}) = \nabla_0 (\nabla_0 \cdot \mathbf{u}) - (\nabla_0 \times \nabla_0 \times \mathbf{u}), \quad (3.65)$$

we can write

$$\nabla_0 \cdot \mathbf{P}_{\text{LIN}}(\boldsymbol{\epsilon}) = \underbrace{(\lambda + 2\mu) \nabla_0 (\nabla_0 \cdot \mathbf{u})}_{\text{dilatational}} - \underbrace{\mu (\nabla_0 \times \nabla_0 \times \mathbf{u})}_{\text{rotational}} \quad (3.66)$$

Finally, we make another assumption

- **Assumption 5:** The **body force** is **negligible**, thus $\mathbf{b}_0 = \mathbf{0}$.

This final assumption allows the **progressive wave equation for solids** to be written as

$$\rho_0 \ddot{\mathbf{u}} = \underbrace{(\lambda + 2\mu) \nabla_0 (\nabla_0 \cdot \mathbf{u})}_{\text{dilatational}} - \underbrace{\mu (\nabla_0 \times \nabla_0 \times \mathbf{u})}_{\text{rotational}} \quad (3.67)$$

There are two wave speeds: the longitudinal sound speed c_L and the transverse sound speed c_T .

$$c_L = \sqrt{\frac{\lambda + 2\mu}{\rho_0}} \quad (\text{longitudinal or P-wave}), \quad (3.68)$$

$$c_T = \sqrt{\frac{\mu}{\rho_0}} \quad (\text{transverse or S-wave}). \quad (3.69)$$

The longitudinal sound speed is faster than the transverse sound speed.

The bulk modulus B is related to the Lamé constants λ and μ through

$$B = \lambda + \frac{2\mu}{3}. \quad (3.70)$$

Substituting for B in the longitudinal sound speed gives

$$c_L = \sqrt{\frac{B + \frac{4}{3}\mu}{\rho_0}} \quad (\text{longitudinal or P-wave}). \quad (3.71)$$

3.9 Plasticity

We first focus on plasticity in one dimension, as opposed to fully three-dimensional plasticity, to emphasize concepts while being as mathematically simple as possible. After the concepts are clear, the extension of the mathematics from one- to three-dimensions can be made naturally.

3.9.1 Perfect Plasticity

The extension of elasticity to plasticity is made by way of *perfect* plasticity. Conceptually, perfect plasticity in the constitutive stress-strain relationship is obtained by “capping” the elastic response function at some yield stress (to be defined later as σ_y). Beyond the yield stress, the stress response function is perfectly horizontal. Increases in strain produce no increase in stress beyond the yield stress.

Deformation Measures

Let the body Ω have strain measure $\varepsilon : \Omega \rightarrow \mathbb{R}$ in one dimension be defined as a ratio of change in length, that is the difference between the new (deformed, current configuration) length ℓ of body Ω and the original (undeformed, reference configuration) length L , divided by the original length,

$$\varepsilon \triangleq \frac{\ell - L}{L}. \quad (3.72)$$

The total strain ε in the body be defined as an additive decomposition of elastic strain ε^e and plastic strain ε^p ,

$$\varepsilon \triangleq \varepsilon^e + \varepsilon^p. \quad (3.73)$$

Constitutive Law

Then, the traditional stress-strain relationship, $\sigma = E\varepsilon^e$, can be rewritten as

$$\sigma = E(\varepsilon - \varepsilon^p). \quad (3.74)$$

A change in the configuration of the plastic element (see Fig. 3.8) can only occur if the notion of **slip rate** is allowed.

That is, *assume* that $\varepsilon^p(t)$ is a function of time on interval $[0, T] \subset \mathbb{R}$. Then from (3.73) and (3.74), time evolution expressions $\varepsilon(t)$ and $\sigma(t)$ exist, respectively.

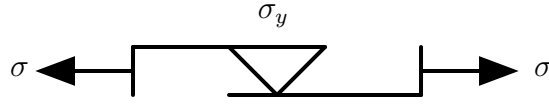


Figure 3.8: Phenomenological model of a friction element. A unit cross-sectional area is assumed. Motion in the friction element occurs only if the friction force, σ_y , is exceeded. The motion, in rate form, is referred to as the slip rate. Note that we employ a slight abuse of notation by using σ to describe a force F , since the cross-sectional area A is unity, $F = \sigma A \implies \sigma \cdot 1 = \sigma$.

Finally, the presence of evolution equations allows the constitutive law in (3.74) to be written as

$$\dot{\sigma} = E (\dot{\varepsilon} - \dot{\varepsilon}^p). \quad (3.75)$$

This form of the constitutive equation will become useful for quantifying plastic flow, as seen below in (3.84).

Yield Function

The stress σ in the frictional device can never greater in absolute value than the yield stress (also known as flow stress) $\sigma_y \in \mathbb{R}^+$. The **yield function** Φ is then defined as

$$\Phi(\sigma) \triangleq |\sigma| - \sigma_y \leq 0. \quad (3.76)$$

For stress levels within the elastic domain, only elastic straining can occur, thus

$$\text{If } \Phi(\sigma) < 0 \Rightarrow \dot{\varepsilon}^p = 0. \quad (3.77)$$

For stress levels on the boundary of the elastic domain (at the yield stress), either plastic strain via plastic loading or non-plastic strain via elastic unloading occurs.

$$\begin{aligned} \text{If } \Phi(\sigma) = 0 \\ \Rightarrow \begin{cases} \dot{\varepsilon}^p \neq 0 & \text{for plastic loading,} \\ \dot{\varepsilon}^p = 0 & \text{for elastic unloading.} \end{cases} \end{aligned} \quad (3.78)$$

Plastic Flow Rule, Loading/Unloading Conditions

We next define how, on the yield surface, the plastic strain evolves through time, which is called the **plastic flow rule**. A reasonable first approximation for the plastic strain evolution is to posit

$$\dot{\epsilon}^p \triangleq \dot{\gamma} \text{sign}(\sigma), \quad (3.79)$$

where $\dot{\gamma} \geq 0$ is the **plastic multiplier** or absolute value of the plastic slip rate, and where $\text{sign} : \mathbb{R} \rightarrow \{-1, +1\}$ is the *signum* function defined as

$$\text{sign}(a) = \begin{cases} +1 & \text{if } a \geq 0, \\ -1 & \text{if } a < 0. \end{cases} \quad (3.80)$$

Both σ and $\dot{\gamma}$ are restricted by **unilateral constraints**,

$$\begin{aligned} \Phi(\sigma) &\leq 0 \\ \dot{\gamma} &\geq 0 \end{aligned} \quad (3.81)$$

Now if $\Phi(\sigma) < 0$, then $\dot{\gamma} = 0$. Similarly, if $\dot{\gamma} > 0$, then $\Phi(\sigma) = 0$. These two observations require

$$\dot{\gamma} \Phi(\sigma) = 0, \quad (3.82)$$

which is referred to as the **complementarity condition**, and when taken with the unilateral constraints, describes the so-called **Kuhn-Tucker conditions**. The unilateral constraints and the complementarity condition are illustrated in Fig. 3.9.

Together, the unilateral constraints and complementarity condition create the normal plasticity law, shown in Fig. 3.10.

Next, consider any time interval $[t, t + \Delta t]$ during which *plastic flow occurs*, the yield function remains constant, $\Phi(\sigma) = 0$. Similarly, if during this time interval, we have elastic response only, and $\dot{\Phi}(\sigma) < 0$, then necessarily $\dot{\gamma} = 0$. These relationships create the **persistency (or consistency) condition**

$$\left. \begin{aligned} \text{If } \dot{\gamma} > 0 &\Rightarrow \dot{\Phi} = 0, \\ \text{If } \dot{\Phi} < 0 &\Rightarrow \dot{\gamma} = 0, \end{aligned} \right\} \Rightarrow \dot{\gamma} \dot{\Phi} = 0. \quad (3.83)$$

Determination of Plastic Multiplier

As of yet, the value of the plastic multiplier $\dot{\gamma}$ has remained undetermined. For non-trivial $\dot{\gamma}$, we know $\dot{\Phi} = 0$ by the persistency condition. An explicit form for $\dot{\Phi}$ may

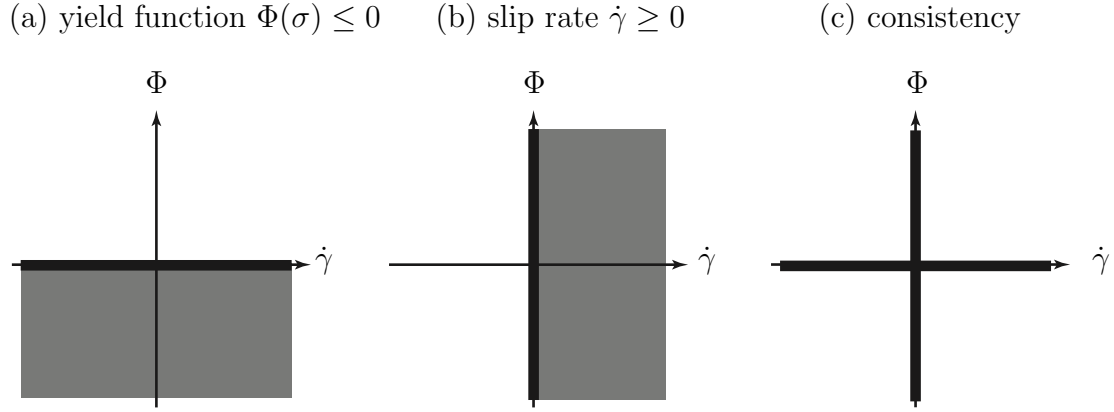


Figure 3.9: Unilateral conditions (a) $\Phi(\sigma) \leq 0$ and (b) slip rate $\dot{\gamma} \geq 0$. The complementarity condition, $\dot{\gamma} \Phi(\sigma) = 0$, is shown in (c).

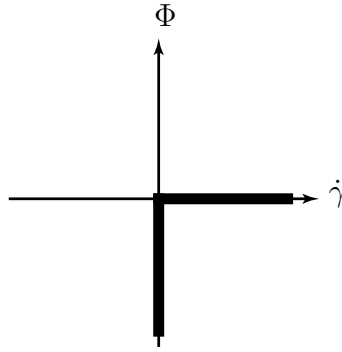


Figure 3.10: The normal plastic law described as the intersection of the unilateral constraints and the complementarity condition.

be found using the chain rule with (3.75) and (3.79),

$$\begin{aligned}
 \frac{\partial \Phi}{\partial t} &= \frac{\partial \Phi}{\partial \sigma} \frac{\partial \sigma}{\partial t} \\
 &= \frac{\partial}{\partial \sigma} (|\sigma| - \sigma_y) E (\dot{\varepsilon} - \dot{\varepsilon}^P) \\
 &= \text{sign}(\sigma) E (\dot{\varepsilon} - \dot{\gamma} \text{sign}(\sigma)),
 \end{aligned} \tag{3.84}$$

or, compactly

$$\dot{\Phi} = \text{sign}(\sigma) E \dot{\varepsilon} - E \dot{\gamma} \stackrel{\text{set}}{=} 0, \tag{3.85}$$

gives

$$\dot{\gamma} = \dot{\varepsilon} \operatorname{sign}(\sigma) \quad (3.86)$$

which, when compared with (3.79), shows that for perfect plasticity during plastic flow,

$$\varepsilon = \varepsilon^P. \quad (3.87)$$

The results from (3.86) and (3.87) are remarkable for the following reasons: During plastic flow,

- None of the strain ε is *elastic* strain ε^e ; 100% of the strain ε is *plastic* strain ε^P . This feature will not be true, in general, for more sophisticated yield functions (for example, that have plastic hardening).
- The plastic flow parameter $\dot{\gamma}$ is equal to the strain ε times the sign of the stress. For more sophisticated yield functions, the plastic flow parameter will not be identically equal to the signed strain, but will be related thereto via factors of the elastic and hardening parameters, as shown subsequent developments and in particular in (3.94).

3.9.2 Hardening Plasticity

Hardening plasticity is considered an extension of perfect plasticity. With perfect plasticity, the slope of stress-strain relationship at yielding is identically zero. With hardening plasticity, the slope becomes positive.⁵

Remarks on Loss of Stability

Perfect plasticity allows for the notion of unbounded strains when stress is at the yield stress. The slope of the stress-strain curve for strains beyond the elastic strain region is identically zero. This zero slope causes numerical problems for the consistent tangent, which is inverted and multiplied by the external load to obtain displacements. Three strategies to circumvent this zero tangent are mentioned here: arc-length, displacement control, and dynamic formulation.

The most notable of the continuation methods is called the **arc-length method**, which is a non-trivial area of study within computational mechanics. Arc-length

⁵For completeness, we note that softening plasticity occurs when the slope becomes negative.

methods will be used eventually, but their explication is delayed for now, in favor of development of hardening plasticity.

Despite being slightly more complex than perfect plasticity, hardening plasticity does not likewise suffer from loss of ellipticity and therefore is actually easier to solve since algorithmic enhancements, such as arc-length, are (usually⁶) unnecessary.

Next, algorithmic formulations that utilize **displacement control** instead of force control can avoid loss of ellipticity if the displacement boundary conditions reduce the number of equations in the tangent to provide rank sufficiency. However, displacement control is less-favored than force control since we often know imposed forces and seek to discover the displacement response. The setting of known inhomogeneous displacements and unknown forces is less frequently encountered in practice. On the topic of load control versus displacement control, de Borst *et al.*,⁷ offer similar insights and additional context.

Finally, elastoplastic **dynamics**, and so-called dynamic relaxation as a subset thereof, are additional strategies to circumvent the above-mentioned problem as these approaches condition the consistent tangent with mass matrices. The benefit of mass matrices comes at the cost of increased expense to compute and store the same, and to incorporate a time stepping algorithm, such as the Newmark methods.

Yield Function

The departure of hardening plasticity from perfect plasticity begins with the yield function. The **yield function** from perfect plasticity in (3.76) is now modified to include a **linear hardening modulus** $H \in \mathbb{R}$ and the **internal hardening variable** $\alpha : [0, T] \rightarrow \mathbb{R}$,

$$\Phi(\sigma, \varepsilon^p) \triangleq |\sigma| - (\sigma_y + H\alpha) \leq 0. \quad (3.88)$$

for strain-hardening plasticity. The variable $\alpha : [0, T] \rightarrow \mathbb{R}$ is called the **internal hardening variable** and is a nonnegative function of the amount of plastic flow

⁶If the hardening is “soft” enough that the hardening approximates perfect plasticity, hardening plasticity can result in lack of conditioning. Usually, however, hardening is sufficiently stiff and therefore loss of ellipticity is not a concern.

⁷De Borst, R., Crisfield, M. A., Remmers, J. J., & Verhoosel, C. V. (2012). Nonlinear finite element analysis of solids and structures. John Wiley & Sons, at page 50–51.

(slip). The simplest evolution for α can be written as

$$\dot{\alpha} = |\dot{\varepsilon}^P|. \quad (3.89)$$

The **plastic flow rule**,

$$\dot{\varepsilon}^P = \dot{\gamma} \text{sign}(\sigma), \quad (3.90)$$

remains unchanged. The **Kuhn-Tucker conditions** are rewritten to include dependence on α ,

$$\dot{\gamma} \geq 0 \quad \Phi(\sigma, \alpha) \leq 0 \quad \dot{\gamma} \Phi(\sigma, \alpha) = 0, \quad (3.91)$$

and the **persistence condition** is written as

$$\dot{\gamma} \dot{\Phi}(\sigma, \alpha) = 0. \quad (3.92)$$

Determination of Plastic Multiplier

Like perfect plasticity, strain-hardening plasticity uses the time derivative of the yield function to generate an expression for the plastic multiplier $\dot{\gamma}$, which ultimately leads to a relationship between stress rates and strain rates, and the elasto-plastic tangent.

With the yield function in (3.88),

$$\begin{aligned} \frac{\partial \Phi}{\partial t} &= \frac{\partial \Phi}{\partial \sigma} \frac{\partial \sigma}{\partial t} + \frac{\partial \Phi}{\partial \alpha} \frac{\partial \alpha}{\partial t}, \\ &= \text{sign}(\sigma) \dot{\sigma} - H \dot{\alpha}, \\ &= \text{sign}(\sigma) E (\dot{\varepsilon} - \dot{\varepsilon}^P) - H |\dot{\varepsilon}^P|, \\ &= \text{sign}(\sigma) E \dot{\varepsilon} - E \dot{\gamma} - H \dot{\gamma} \stackrel{\text{set}}{=} 0, \end{aligned} \quad (3.93)$$

gives

$$\boxed{\dot{\gamma} = \dot{\varepsilon} \text{sign}(\sigma) \frac{E}{E + H}}. \quad (3.94)$$

The ratio $\frac{E}{E+H}$ in (3.94) may be conceived as the apportionment of total strain (rate) to plastic slip:

- If $H = 0$, then all strain is plastic slip, and the perfect plasticity results obtained in previous sections are obtained as a special case.
- If $H = E$, then half of all strain goes into plastic slip.
- If $H \gg E$, then very little $\dot{\varepsilon}$ goes into plastic slip.

Elasto-Plastic Tangent

The elasto-plastic tangent is found by observing $\dot{\sigma} = \frac{d\sigma}{d\varepsilon}\dot{\varepsilon}$. Starting with the elastic stress-strain relationship $\sigma = E(\varepsilon - \varepsilon^p)$, in the rate form and substituting for $\dot{\varepsilon}$ from (3.94) and for $\dot{\varepsilon}^p$ from the plastic flow rule (3.90) gives

$$\begin{aligned}\dot{\sigma} &= E(\dot{\varepsilon} - \dot{\varepsilon}^p), \\ &= \dot{\gamma} \operatorname{sign}(\sigma)(E + H) - \dot{\gamma} \operatorname{sign}(\sigma)E, \\ &= \dot{\gamma} \operatorname{sign}(\sigma)H.\end{aligned}\tag{3.95}$$

Substituting for $\dot{\gamma}$ from (3.94) gives

$$\dot{\sigma} = \frac{EH}{E + H}\dot{\varepsilon},\tag{3.96}$$

where E_{ep} , the elasto-plastic tangent, is giving by

$$\boxed{E_{ep} = \frac{EH}{E + H}}.\tag{3.97}$$

The concept of the elasto-plastic tangent is readily understood with the phenomenological model shown in Fig. 3.11.

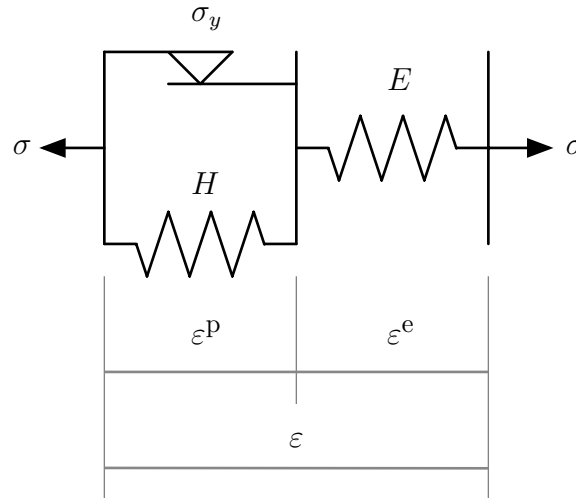


Figure 3.11: Phenomenological model of strain-hardening plasticity with applied stress σ , yield stress σ_y , hardening modulus H and elastic modulus E . The total strain, ε is the sum of plastic strain ε^p and elastic strain ε^e .

The stress σ is applied to the model. Initially, the applied stress σ creates only elastic strain ε^e since $\sigma \leq \sigma_y$. When $\sigma > \sigma_y$, strain composed of *both* elastic strain ε^e and plastic strain ε^p is created. The linear moduli E and H act as springs in series. The relationship for two springs with spring constants, k_1 and k_2 , in series immediately shows the relationship between the equivalent spring constant k_{eq} and the elasto-plastic tangent E_{ep}

$$\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2} \implies k_{eq} = \frac{k_1 k_2}{k_1 + k_2} \quad (3.98)$$

$$\begin{array}{ccc} \Downarrow & & \Downarrow \\ E_{ep} & = & \frac{EH}{E + H} \end{array} \quad (3.99)$$

The hardening modulus H is shown in Fig. 3.12(b) such that, during plastic flow, the incremental stress $\partial\sigma$ relative to the incremental plastic strain $\partial\varepsilon^p$ is given by

$$\frac{\partial\sigma}{\partial\varepsilon^p} = H. \quad (3.100)$$

3.9.3 Elasto-Plastic 2D Truss Element

Consider the two-dimensional truss member shown in Fig. 3.13. The coordinates in the reference configuration are noted with $\mathbf{X}^1$ and $\mathbf{X}^2$ for nodes $\dot{1}$ and $\dot{2}$, respectively. These nodes are moved into the current configuration at $\mathbf{x}^1$ and $\mathbf{x}^2$ through nodal displacements $\mathbf{u}^1$ and $\mathbf{u}^2$.

The strain energy of the truss is the strain energy density integrated over the truss's volume,

$$\mathcal{W} = \int_{\Omega} \boldsymbol{\sigma} : \boldsymbol{\varepsilon} \, d\Omega. \quad (3.101)$$

The virtual work done by the truss is

$$\delta\mathcal{W} = \int_{\Omega} \delta\boldsymbol{\varepsilon} : \boldsymbol{\sigma} \, d\Omega. \quad (3.102)$$

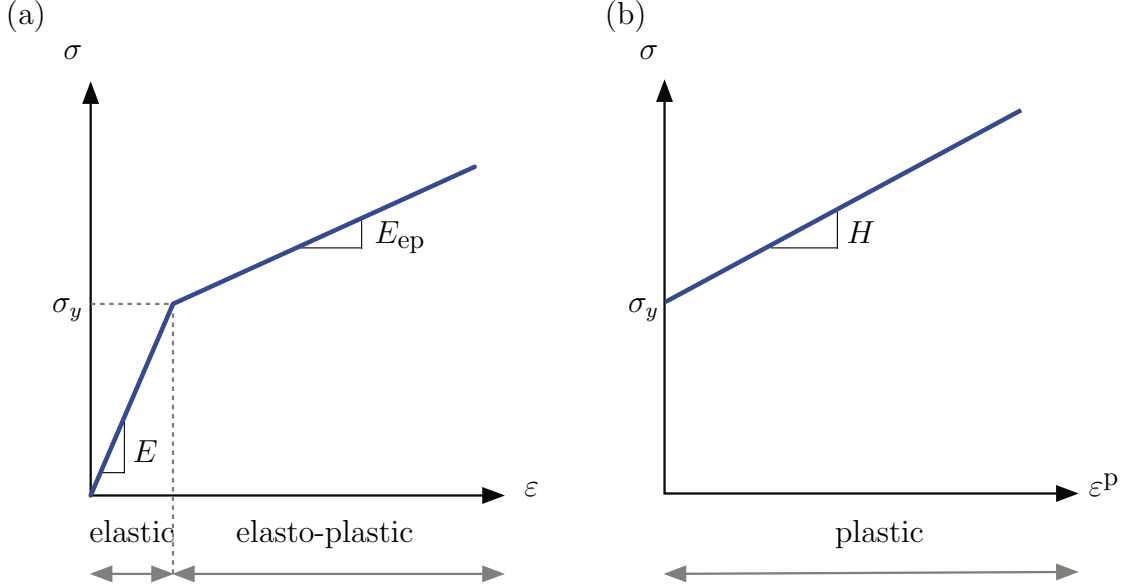


Figure 3.12: (a) Strain-hardening stress-strain relationship with elastic and elasto-plastic strain regions delineated, elastic modulus E , elasto-plastic modulus E_{ep} , and yield stress σ_y , and (b) stress-(plastic) strain relationship with hardening modulus H .

The virtual work computed in the reference configuration is then

$$\delta \mathcal{W} = \int_L \delta \boldsymbol{\epsilon} : \boldsymbol{\sigma} A \, dL, \quad (3.103)$$

$$\stackrel{(1D)}{=} \delta \left(\frac{\ell - L}{L} \right) \sigma A L = \frac{\delta \ell}{L} \sigma A L = \delta \ell \sigma A, \quad (3.104)$$

$$\stackrel{(2.70)}{=} (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \cdot \hat{\mathbf{t}} \sigma A, \quad (3.105)$$

$$\stackrel{(\text{lin})}{=} (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \cdot \hat{\mathbf{t}} E \left(\frac{\ell - L}{L} \right) A, \quad (3.106)$$

$$= (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \cdot \hat{\mathbf{t}} k(\ell - L), \quad (3.107)$$

where the result from Hovey⁸ page 45, Eq. (2.70), $\delta \ell = (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \cdot \hat{\mathbf{t}}$, has been used, and agrees with Eq. (2.76) of the same manuscript. The virtual work expressed

⁸Hovey CB (2000). Stress and Motion in Diarthrodial Joints Using Coupled Nonlinear Elastodynamics, Rigid Body Dynamics, and Large-Slip Contact-Impact, Doctoral Thesis, Stanford University, Stanford, CA.

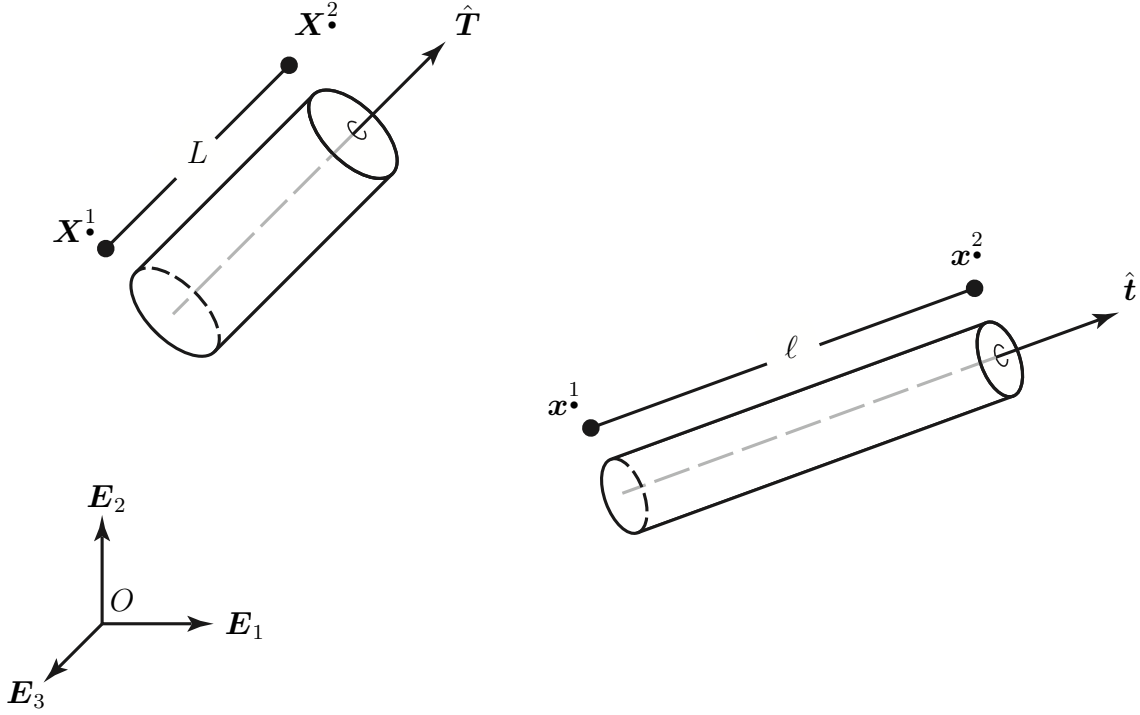


Figure 3.13: Nonlinear truss in reference and current configurations.

in terms of nodal displacements and the *nonlinear* internal force vector in (3.105) is then

$$\delta\mathcal{W} = \boldsymbol{\delta} \cdot \mathbf{f}^{\text{int}} = \left\{ \begin{array}{c} \delta \mathbf{u}^1 \\ \delta \mathbf{u}^2 \end{array} \right\}^T \sigma A \left\{ \begin{array}{c} -\hat{\mathbf{t}} \\ \hat{\mathbf{t}} \end{array} \right\}. \quad (3.108)$$

With $\delta\ell$ at hand, and noting that

$$\delta\hat{\mathbf{t}} = \delta \left(\frac{\mathbf{x}_2 - \mathbf{x}_1}{l} \right) = \delta \left((\mathbf{x}_2 - \mathbf{x}_1) l^{-1} \right), \quad (3.109)$$

$$= \frac{1}{l} [\mathbf{1} - \hat{\mathbf{t}} \otimes \hat{\mathbf{t}}] (-\delta \mathbf{u}_1 + \delta \mathbf{u}_2), \quad (3.110)$$

the second variation of the nonlinear form in (3.105), or equivalently in (3.108), is found through

$$\Delta(\delta\mathcal{W}) = \Delta \left[(-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \cdot \sigma A \hat{\mathbf{t}} \right], \quad (3.111)$$

$$\begin{aligned} &= (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \cdot \Delta \sigma A \hat{\mathbf{t}} \\ &\quad + (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \cdot \sigma A \Delta \hat{\mathbf{t}}. \end{aligned} \quad (3.112)$$

Now, adopting a compact notation where $\delta \mathbf{u} = (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2)$ and $\Delta \mathbf{u} = (-\Delta \mathbf{u}^1 + \Delta \mathbf{u}^2)$,

$$\begin{aligned} \Delta(\delta \mathcal{W}) &= \delta \mathbf{u} \cdot \left\{ \frac{\partial \sigma}{\partial \varepsilon} \Delta \varepsilon A \hat{\mathbf{t}} + \sigma A \frac{1}{\ell} [\mathbf{1} - \hat{\mathbf{t}} \otimes \hat{\mathbf{t}}] \Delta \mathbf{u} \right\}, \end{aligned} \quad (3.113)$$

$$= \delta \mathbf{u} \cdot \left\{ \frac{\partial \sigma}{\partial \varepsilon} \left(\frac{\Delta \ell}{L} \right) A \hat{\mathbf{t}} + \frac{\sigma A}{\ell} [\mathbf{1} - \hat{\mathbf{t}} \otimes \hat{\mathbf{t}}] \Delta \mathbf{u} \right\}. \quad (3.114)$$

Expanding $\delta \mathbf{u}$ and $\Delta \mathbf{u}$ gives

$$\begin{aligned} \Delta(\delta \mathcal{W}) &= (-\delta \mathbf{u}^1 + \delta \mathbf{u}^2) \\ &\quad \cdot \left[\frac{\partial \sigma}{\partial \varepsilon} \frac{A}{L} \hat{\mathbf{t}} \otimes \hat{\mathbf{t}} + \frac{\sigma A}{\ell} [\mathbf{1} - \hat{\mathbf{t}} \otimes \hat{\mathbf{t}}] \right] \\ &\quad \left\{ (-\Delta \mathbf{u}^1 + \Delta \mathbf{u}^2) \right\}. \end{aligned} \quad (3.115)$$

This leads to the element stiffness equations

$$\begin{aligned} \Delta \delta \mathcal{W} &= \\ \delta \cdot \mathbf{K} \Delta &= \begin{Bmatrix} \delta \mathbf{u}^1 \\ \delta \mathbf{u}^2 \end{Bmatrix}^T \begin{bmatrix} \boldsymbol{\kappa} & -\boldsymbol{\kappa} \\ -\boldsymbol{\kappa} & \boldsymbol{\kappa} \end{bmatrix} \begin{Bmatrix} \Delta \mathbf{u}^1 \\ \Delta \mathbf{u}^2 \end{Bmatrix}, \end{aligned} \quad (3.116)$$

where

$$\boldsymbol{\kappa} = \left[\frac{\partial \sigma}{\partial \varepsilon} \frac{A}{L} \hat{\mathbf{t}} \otimes \hat{\mathbf{t}} + \frac{\sigma A}{\ell} [\mathbf{1} - \hat{\mathbf{t}} \otimes \hat{\mathbf{t}}] \right]. \quad (3.117)$$

Note that the explicit form of $\partial \sigma / \partial \varepsilon$ will depend on the plastic multiplier $\dot{\gamma}$. If $\dot{\gamma} = 0$, then $\partial \sigma / \partial \varepsilon = E$, as shown in the elastic region of Fig. 3.12 (a). If $\dot{\gamma} > 0$, then $\partial \sigma / \partial \varepsilon = E_{\text{ep}}$, as shown in the elasto-plastic region of Fig. 3.12 (a).

3.9.4 From 1D to 3D

In one-dimensional plasticity, we spoke of the Cauchy stress scalar $\sigma \in \mathbb{R}$ with positive values describing tension (resulting in elongations) and compression (resulting in contractions). Now, we generalize this concept to speak of the Cauchy stress tensor $\boldsymbol{\sigma}$ as a collection of all the normal and shear Cauchy stress components present in three dimensions.

We decompose $\boldsymbol{\sigma}$ into volumetric and deviatoric components, denoted

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}_{\text{VOL}} + \boldsymbol{\sigma}_{\text{DEV}}. \quad (3.118)$$

Next, we *define* each of these two components. The **volumetric** Cauchy stress tensor is defined as

$$\boldsymbol{\sigma}_{\text{VOL}} \triangleq \frac{1}{3} \text{tr}(\boldsymbol{\sigma}) \mathbf{1} = \frac{1}{3} \sigma_{kk} \delta_{ij}. \quad (3.119)$$

It is often convenient to define **pressure** $P \in \mathbb{R}$ as

$$P \triangleq -\frac{1}{3} \text{tr}(\boldsymbol{\sigma}) \quad \begin{cases} P > 0 \text{ compression,} \\ P < 0 \text{ tension.} \end{cases} \quad (3.120)$$

Note the negative sign in the definition. The negative sign allows us to harmonize our concepts of (compressive, e.g., underwater) pressure as a positive, scalar with tension/compression (positive/negative) convention of the Cauchy stress tensor.

Furthermore, the deviatoric Cauchy stress tensor is often denoted simply as $\mathbf{s}$. Using this notation, the **deviatoric** Cauchy stress tensor is defined as

$$\mathbf{s} \triangleq \boldsymbol{\sigma} - \boldsymbol{\sigma}_{\text{VOL}} = \boldsymbol{\sigma} + P \mathbf{1}. \quad (3.121)$$

Uniaxial Tension Test

With these definitions at hand, we are now poised to demonstrate how the results of a uniaxial tension test at the plastic limit are used to generalize a theory of small-strain plasticity to three dimensions.

A yielding, the Cauchy stress tensor $\boldsymbol{\sigma}$ has the simple form

$$\boldsymbol{\sigma}|_y = \begin{bmatrix} \sigma_y & 0 & 0 \\ & 0 & 0 \\ & & 0 \end{bmatrix}. \quad (3.122)$$

Remark 3.9.1. Note that for symmetric matrices, we will avoid filling in the lower part of the matrix both for brevity, and to reinforce the idea of symmetry.

The volumetric part of the Cauchy stress tensor at yield is then

$$\boldsymbol{\sigma}_{\text{VOL}}|_y = \frac{\sigma_y}{3} \begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ & & 1 \end{bmatrix}. \quad (3.123)$$

The deviatoric part of the Cauchy stress tensor at yield is then

$$\mathbf{s}|_y = \frac{3\sigma_y}{3} \begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ & & 1 \end{bmatrix} - \frac{\sigma_y}{3} \begin{bmatrix} 1 & 0 & 0 \\ & 1 & 0 \\ & & 1 \end{bmatrix} = \frac{\sigma_y}{3} \begin{bmatrix} 2 & 0 & 0 \\ & -1 & 0 \\ & & -1 \end{bmatrix}. \quad (3.124)$$

So, we now have the form of the deviatoric Cauchy stress tensor $\mathbf{s}$ at yield. How can this result for a uniaxial tension test be generalized for three dimensions? The thought experiment is to envision $\mathbf{s}$ for *any and all* matrix forms, not just the form obtained in (3.124). Any and all matrix form for $\mathbf{s}$ will have one commonality: their matrix invariants.

Recall that the three invariants, $I_1(\mathbf{A})$, $I_2(\mathbf{A})$, and $I_3(\mathbf{A})$ of matrix $\mathbf{A}$ satisfy the equation

$$\det(\mathbf{A} - \lambda \mathbf{1}) \stackrel{\text{set}}{=} 0 \quad (3.125)$$

for any λ , and produce the characteristic equation

$$0 = \lambda^3 (-1) \quad (3.126)$$

$$+ \lambda^2 \underbrace{(a_{11} + a_{22} + a_{33})}_{\text{tr}(\mathbf{A})} \quad (3.127)$$

$$+ \lambda^1 \underbrace{(-a_{11}a_{22} - a_{22}a_{33} - a_{33}a_{11} + a_{12}a_{21} + a_{23}a_{32} + a_{31}a_{13})}_{\frac{1}{2} [\text{tr}(\mathbf{A} \cdot \mathbf{A}) - \text{tr}(\mathbf{A})\text{tr}(\mathbf{A})]} \quad (3.128)$$

$$+ \lambda^0 \underbrace{(a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31})}_{\det(\mathbf{A})}. \quad (3.129)$$

Then the first invariant of $\mathbf{s}$,

$$J_1(\mathbf{s}) \triangleq \text{tr}(\mathbf{s}) = s_{kk} = 0, \quad (3.130)$$

is uninteresting because it has zero value for all stress states of $\boldsymbol{\sigma}$. The second invariant of $\mathbf{s}$,

$$J_2(\mathbf{s}) \triangleq \frac{1}{2} (\text{tr}(\mathbf{s} \cdot \mathbf{s})) = \frac{1}{2} s_{ij} s_{ji} \quad (3.131)$$

is interesting because we arrive at a quadratic form of $\mathbf{s}$. Returning to the thought experiment, we can now envision of form of $\mathbf{s}$ by way of $J_2(\mathbf{s})$ as valid for all values

Cauchy stress tensors $\boldsymbol{\sigma}$ and with plasticity occurring when $J_2(\mathbf{s})$ reaches $J_2(\mathbf{s}|_y)$. Said equivalently, we can imagine a yield function, as defined for scalars in (3.76), recast now for the entire stress tensor $\boldsymbol{\sigma}$ as

$$\Phi(\boldsymbol{\sigma}) \triangleq J_2(\mathbf{s}(\boldsymbol{\sigma})) - J_2(\mathbf{s}|_y) \leq 0. \quad (3.132)$$

Now

$$J_2(\mathbf{s}|_y) = \frac{1}{2} (\sigma_y^2)/9 (4 + 1 + 1) = \sigma_y^2/3. \quad (3.133)$$

Thus

$$\Phi(\boldsymbol{\sigma}) \triangleq J_2(\mathbf{s}(\boldsymbol{\sigma})) - \frac{\sigma_y^2}{3} \leq 0. \quad (3.134)$$

Now, on the yield surface, $\Phi = 0$, which means

$$J_2(\mathbf{s}(\boldsymbol{\sigma}|_y)) = \frac{\sigma_y^2}{3} \implies \sqrt{3J_2(\mathbf{s}(\boldsymbol{\sigma}|_y))} = \sigma_y. \quad (3.135)$$

This gives rise to the form of the yield function of so-called J_2 **plasticity**,

$$\boxed{\Phi(\boldsymbol{\sigma}) \triangleq \sqrt{3J_2(\mathbf{s}(\boldsymbol{\sigma}))} - \sigma_y \leq 0.} \quad (3.136)$$

The quantity $\sqrt{3J_2}$ is often referred to as the **von Mises stress** or **equivalent tensile stress** and denoted as σ_{VM} . Thus, the yield function may be equivalently written as

$$\Phi(\boldsymbol{\sigma}) \triangleq \sigma_{\text{VM}}(\boldsymbol{\sigma}) - \sigma_y \leq 0. \quad (3.137)$$

3.10 One Dimensional Contact

3.10.1 Concepts

Gap and Penetration

Consider two bodies $\mathcal{A}$ and $\mathcal{B}$ in moving in inertial reference from $\mathcal{F}$ with origin O and basis vector $\hat{\mathbf{f}}_1$, as shown in Fig. 3.14. Let point P be located on the right-hand-side of body $\mathcal{A}$, denoting the first contact surface $\partial\mathcal{A}$. Let point Q be located on the left-hand-side of body $\mathcal{B}$, denoting the second contact surface $\partial\mathcal{B}$. Let the contact boundaries be denoted as $\Gamma_c = \partial\mathcal{A} \cup \partial\mathcal{B}$. Let the time evolution from $t = 0$ (initial) to $t = T$ (final) be denoted as $\mathbb{T} = [0, T] \subset \mathbb{R}^+$. At the initial time $t = 0$, these two points are located by X^{OP} and X^{OQ} , respectively.

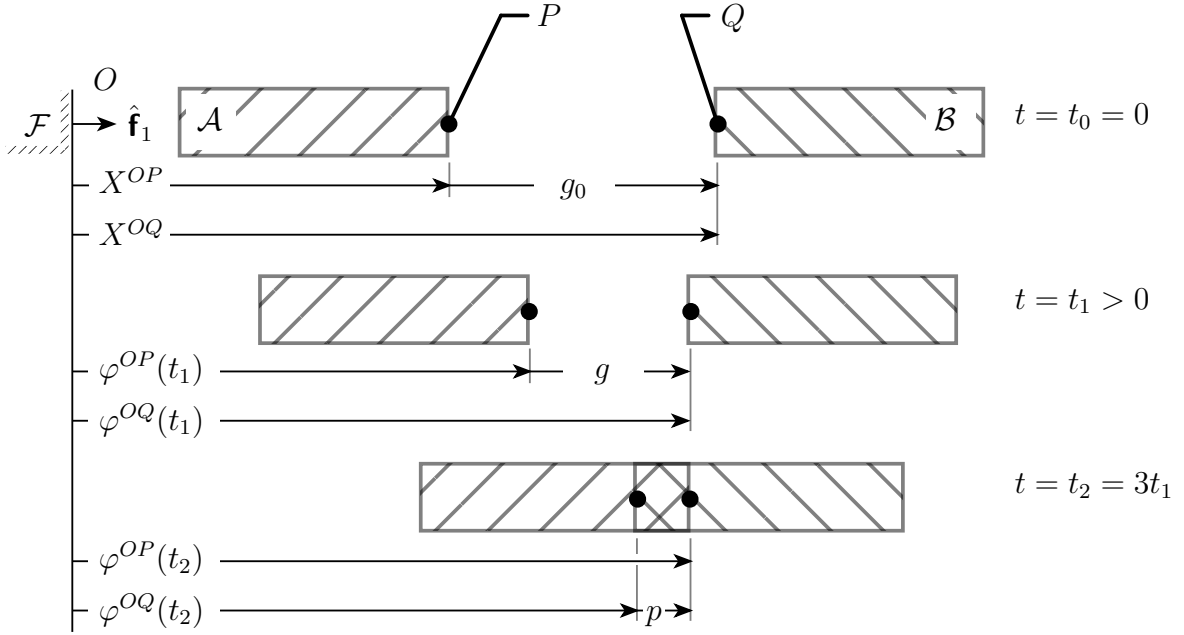


Figure 3.14: Caption to come.

At $t = 0$, the initial gap $g_0 : [\Gamma_c \times 0] \mapsto \mathbb{R}^+$ between the two points P and Q is given by

$$g(t = 0) = g_0 = X^{OQ} - X^{OP} \triangleq X^{PQ}, \quad g_0 > 0. \quad (3.138)$$

In general, the gap $g : [\Gamma_c \times \mathbb{T}] \mapsto \mathbb{R}$ between the points is given by

$$g(\varphi(\mathbf{X}), t) = \varphi_t^{OQ} - \varphi_t^{OP} = (X^{OQ} + u_t^Q) - (X^{OP} + u_t^P) = g_0 + u_t^Q - u_t^P, \quad (3.139)$$

where u_t is a displacement, φ_t is a current configuration, $\mathbf{u} = \langle u_t^P, u_t^Q \rangle^T$, and $\boldsymbol{\varphi} = \langle \varphi_t^{OP}, \varphi_t^{OQ} \rangle^T$. For arbitrary displacements u_t^P and u_t^Q , the gap can be either positive, negative, or zero. For example, at $t = t_2$, the gap is negative. It is useful, as will be seen in subsequent discussions on the penalty formulation (see ??), to define penetration p as the negative of the gap. The penetration $p : [\Gamma_c \times \mathbb{T}] \mapsto \mathbb{R}$ is defined as

$$p(\mathbf{u}, t) \triangleq -g(\mathbf{u}, t) = -\varphi_t^{OQ} + \varphi_t^{OP} = -(X^{OQ} + u_t^Q) + (X^{OP} + u_t^P) = -g_0 - u_t^Q + u_t^P. \quad (3.140)$$

Note that

$$p(t = 0) = p_0 = -g_0, \quad p_0 < 0. \quad (3.141)$$

Non-Penetration

The penetration of body $\mathcal{A}$ and body $\mathcal{B}$, shown (for example) at $t = t_2$ in Fig. 3.14, is prevented by constraining the gap g (or, equivalently, the penetration p) as follows:

$$g \geq 0, \quad (3.142)$$

$$p \leq 0. \quad (3.143)$$

These two inequality conditions are **unilateral constraints** on the motion $\boldsymbol{\varphi}$. To illustrate the unilateral constraint concept in a concrete sense, consider $\mathcal{B}$ in Fig. 3.14 to be stationary for all time. Thus $u_t^Q = 0$, and we can abbreviate the displacement of body $\mathcal{A}$ as $u_t^P = u$. The gap and penetration functions are then

$$g(u) = g_0 - u, \quad (3.144)$$

$$p(u) = -g(u) = u - g_0 = u + p_0. \quad (3.145)$$

Assume, for concreteness $g_0 = 1$ m. The gap and penetration functions are then shown in Fig. 3.15.

3.10.2 Example with Spring and Weight

We start by considering a spring mass system, with spring constant k , mass m , gravitational field $-g \hat{\mathbf{f}}_2$, with the spring initially unstretched at $t = t_0 = 0$, as shown in Fig. 3.16. While this example does not yet have contact, it sets the context of energy minimization as method used to find static equilibrium.

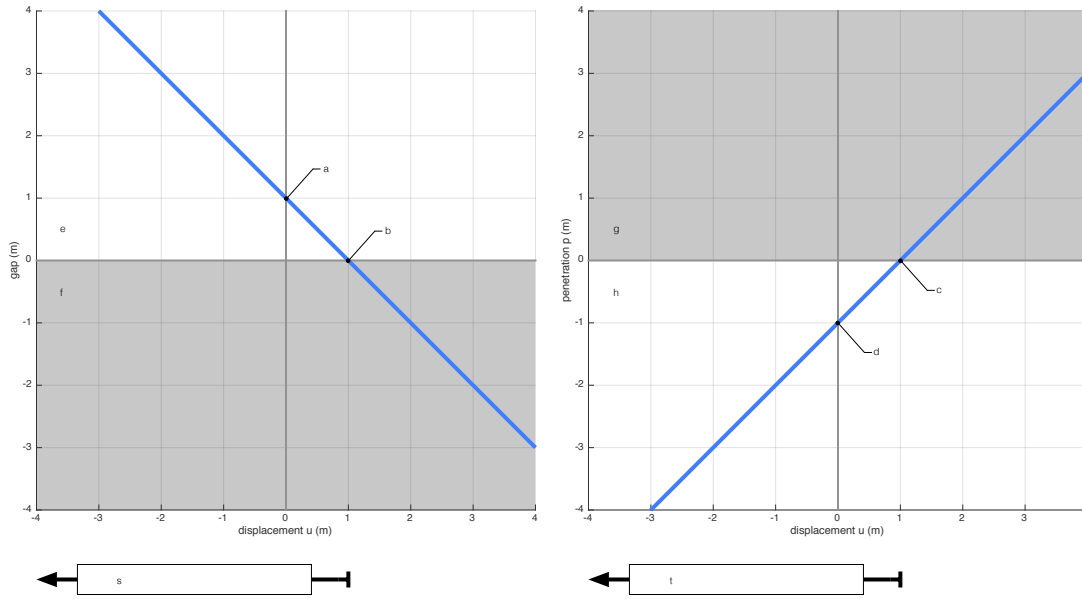


Figure 3.15: Caption to come.

For concreteness, let $k = 2 \text{ N/m}$, $m = 0.2 \text{ kg}$, and $g = 10 \text{ m/s}^2$, (for simplicity, rounded up from 9.81 m/s^2). Let displacement u of the mass, initially at 0, be measured in the $\hat{\mathbf{f}}_2$ axis. The internal energy (IE) of the spring as a function of u is given by $\Pi_k(u) = \frac{1}{2}ku^2$. The potential energy (PE) of the mass as a function of u is given by $\Pi_m(u) = mgu$. Minimization of the potential energy gives

$$\delta\Pi = \delta(\Pi_k(u) + \Pi_m(u)) = \delta\left(\frac{1}{2}ku^2 + mgu\right) = (ku + mg)\delta u \stackrel{\text{set}}{=} 0, \quad (3.146)$$

$$\implies u = \frac{-mg}{k} = \frac{-0.2 \cdot 10}{2} = -1. \quad (3.147)$$

Fig. 3.17 shows the spring internal energy, $\Pi_k(u)$, in dashed blue, which shows a minimum at $u = 0$, and a symmetric, parabolic response for nonzero displacements, as expected. The potential energy due to the mass, $\Pi_m(u)$, is shown in dot-dash green, which shows a linear relationship in u with intercept 0 and slope $mg = 0.2 \cdot 10 = 2 \text{ N}$.

The addition of the two aforementioned potentials, $\Pi(u) = \Pi_k(u) + \Pi_m(u)$, is shown in solid red. The system potential, $\Pi(u)$, has a minimum at $u = -1 \text{ m}$, as expected. Next, we used this concept of minimizing potential energy in the context of contact.

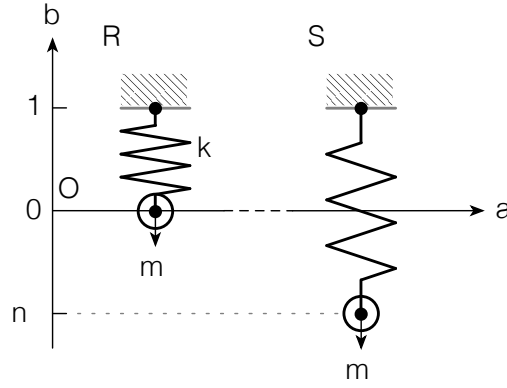


Figure 3.16: Caption to come.

3.10.3 Example with Spring and Rigid Wall

Consider a spring, shown in Fig. 3.18, with elastic constant $k = 2 \text{ N/m}$, undeformed length $L_0 = 2 \text{ m}$ subjected to a prescribed displacement in the amount of $\bar{u} = 2 \text{ m}$ on the left-hand-side.⁹ A rigid wall is $g_0 = X^{PQ} = 1 \text{ m}$ from the right-hand-side of the spring at $t = 0$. At $t = t_1$, the spring has undergone its prescribed displacement and penetrates the wall by amount p . In this configuration, internal energy Π for the spring for the as a function of $u = u^P$ is given by

$$\Pi(u) = \frac{1}{2} k \delta^2 = \frac{1}{2} 2 (u - \bar{u})^2 = (u - 2)^2 (\text{Nm}). \quad (3.148)$$

Principle of Minimum Potential Energy

Principle of Minimum Potential Energy with Lagrange Multiplier Enforcement

⁹This example follows Laursen TA (2013), *Computational contact and impact mechanics: fundamentals of modeling interfacial phenomena in nonlinear finite element analysis*. Springer Science & Business Media, page 87.

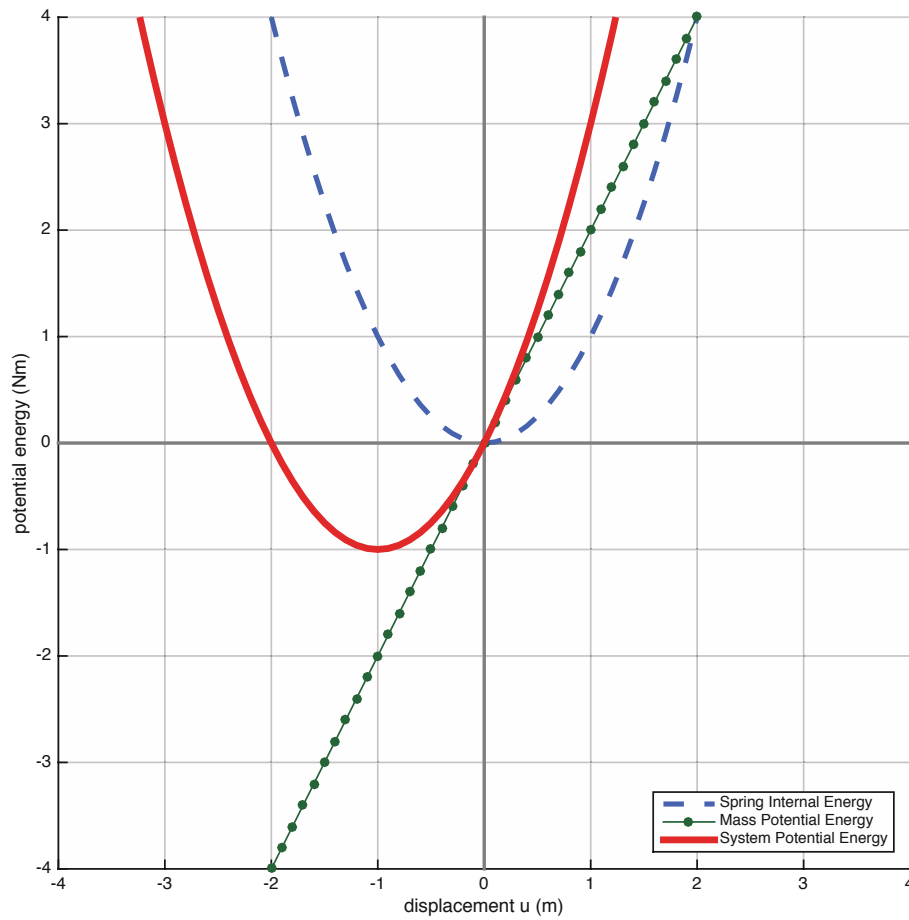


Figure 3.17: Caption to come.

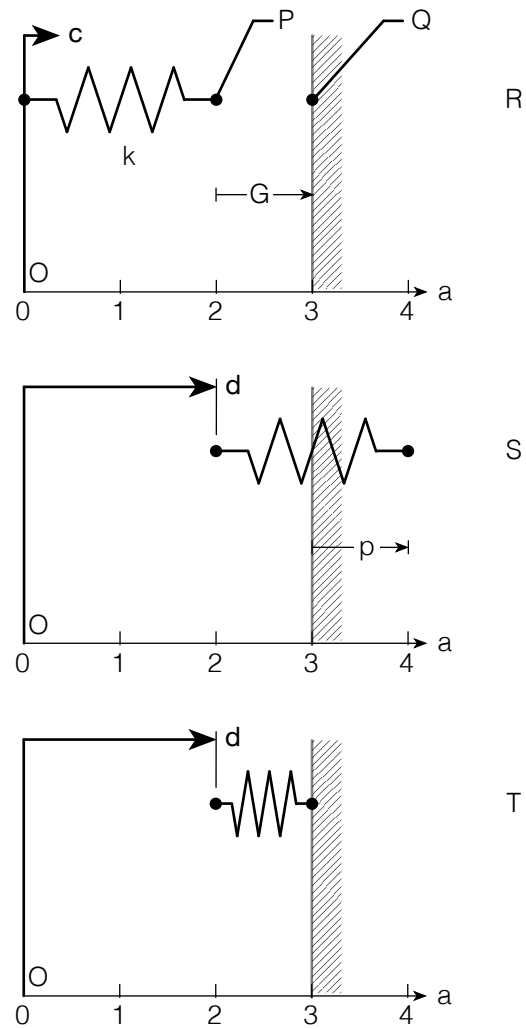


Figure 3.18: Caption to come.

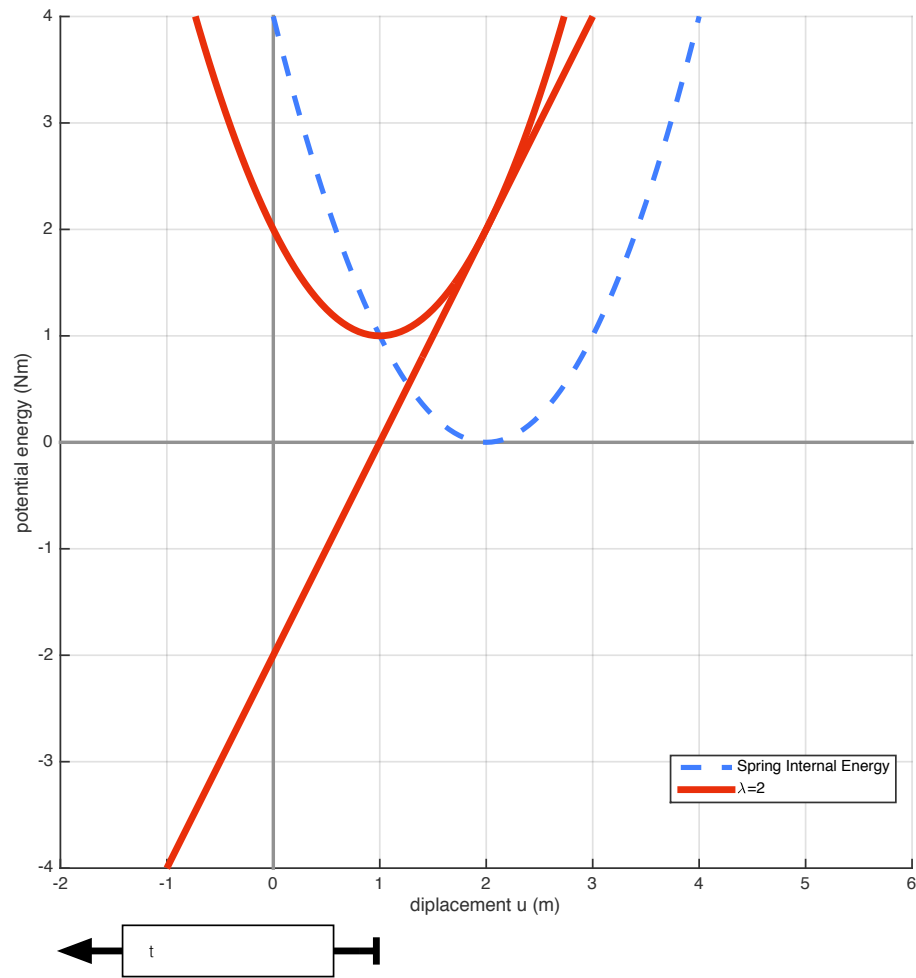


Figure 3.19: Caption to come.

$$\Pi(u) = \Pi_k(u) + \Pi_c(u). \quad (3.149)$$

$$= \frac{1}{2}k(u - \bar{u})^2 + \lambda p(u). \quad (3.150)$$

$$= (u - \bar{u})^2 + \lambda(u + p_0). \quad (3.151)$$

$$\text{where } \bar{u} = 2 \text{ m, and } p_0 = -1 \text{ m.} \quad (3.152)$$

Principle of Minimum Potential Energy with Penalty Enforcement

Principle of Minimum Potential Energy with Augmented Lagrange Multiplier Enforcement

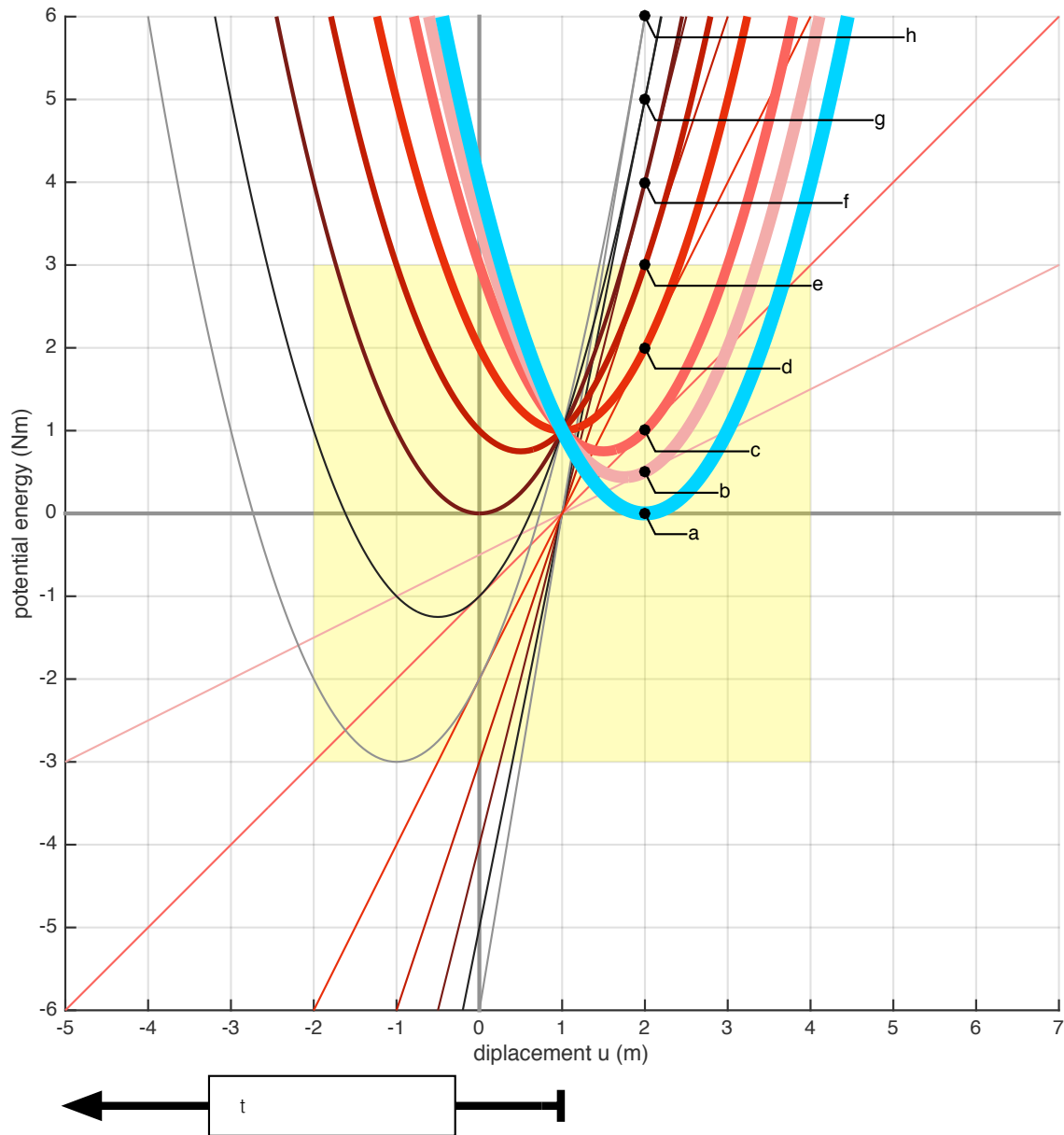


Figure 3.20: Caption to come.

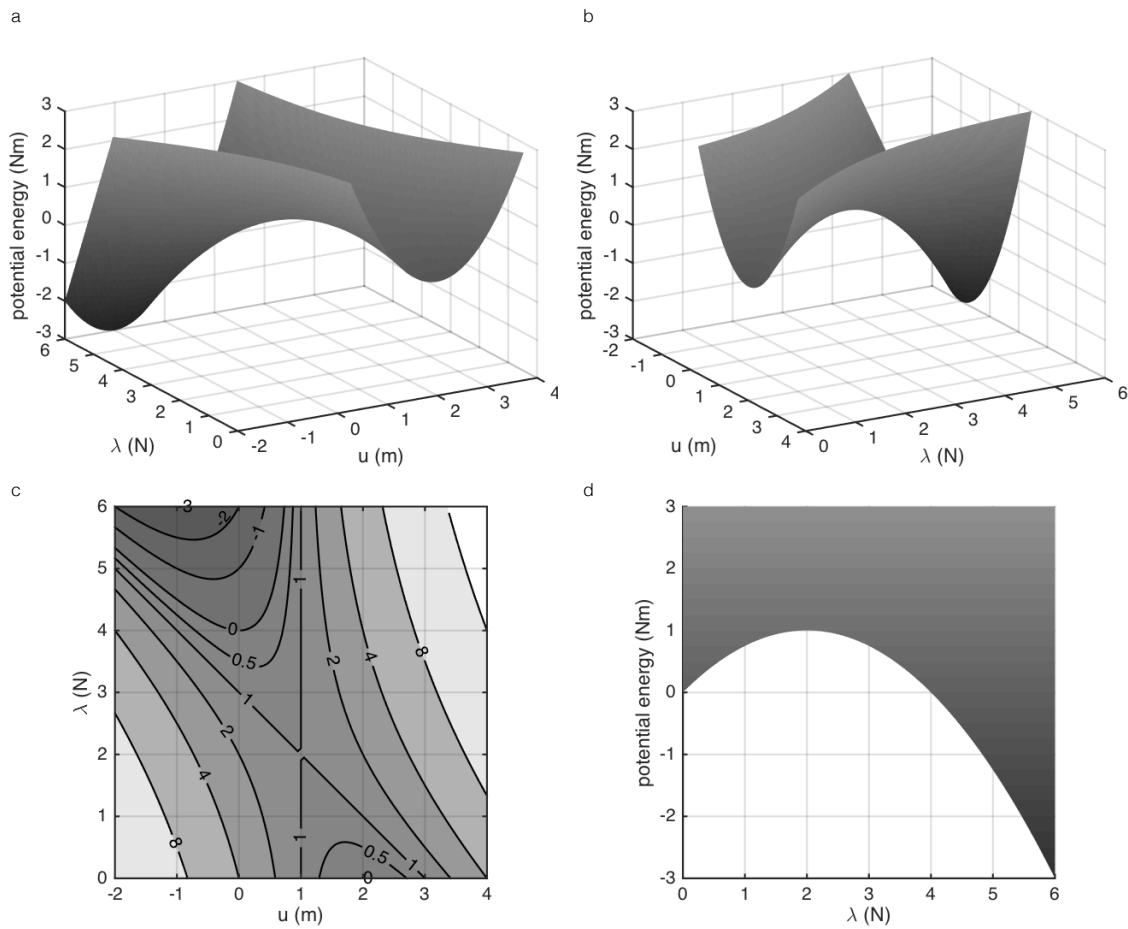


Figure 3.21: Caption to come.